Dynamics and Control of Cranes: A Review

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(Received 5 April 2001; accepted 19 September 2001)

Abstract: We review crane models available in the literature, classify them, and discuss their applications and limitations. A generalized formulation of the most widely used crane model is analyzed using the method of multiple scales. We also review crane control strategies in the literature, classify them, and discuss their applications and limitations. In conclusion, we recommend appropriate models and control criteria for various crane applications and suggest directions for further work.

Key Words: Crane, dynamics, control, stability, gantry crane, rotary crane, boom crane

1. INTRODUCTION

Cranes are increasingly used in transportation and construction. They are also becoming larger, faster, and higher, necessitating efficient controllers to guarantee fast turn-over time and to meet safety requirements. In the last 40 years we have seen mounting interest in research on the modeling and control of cranes. In this paper, we review this body of literature available in the English language journals and conference proceedings.

A crane consists of a hoisting mechanism (traditionally a hoisting line and a hook) and a support mechanism (trolley-girder, trolley-jib, or a boom). The cable–hook–payload assembly is suspended from a point on the support mechanism. The support mechanism moves the suspension point around the crane workspace, while the hoisting mechanism lifts and lowers the payload to avoid obstacles in the path and deposit the payload at the target point.

Cranes can be classified based on the degrees of freedom the support mechanism offers the suspension point. The support mechanism in a *gantry (overhead) crane*, Figure 1, is composed of a trolley moving over a girder. In some gantry cranes, this girder (bridge) is in turn mounted on another set of orthogonal railings in the horizontal plane. This setup allows the suspension point one or two rectilinear translations in the horizontal plane. In a (tower) *rotary crane*, Figure 2, the girder (jib) rotates in the horizontal plane about a fixed vertical axis. This allows the suspension point two motion patterns in the horizontal plane,

Journal of Vibration and Control, **9:** 863–908, 2003 DOI: 10.1177/107754603031852
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Figure 1. A bidirectional gantry crane.

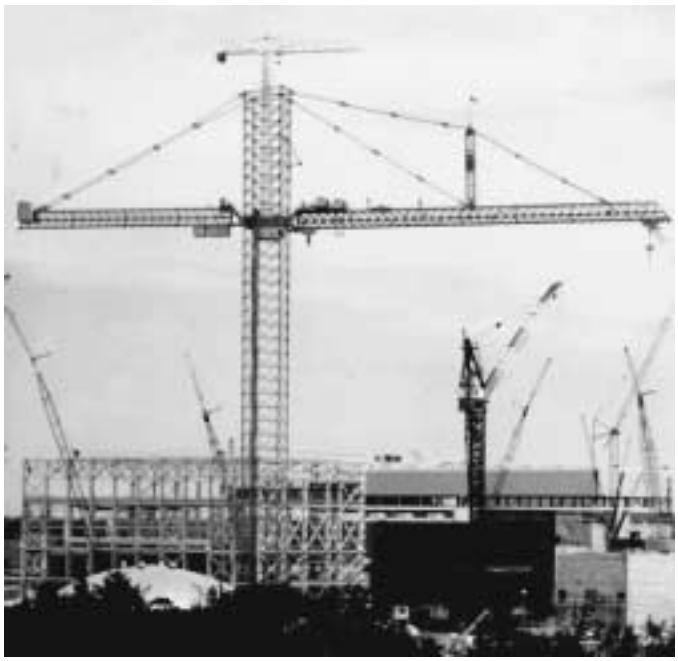


Figure 2. A rotary crane.



Figure 3. A boom crane.

a translation and a rotation. The suspension point in a *boom crane*, Figure 3, is fixed at the end of the boom. It has two motion patterns: rotations around two orthogonal axes located at the base of the boom.

The capacity of the boom to support loads in compression (as opposed to bending) offers boom cranes a structural advantage over other types of crane. As a result, boom cranes are compact in comparison to similar capacity gantry or rotary cranes. Consequently, all mobile cranes use boom cranes. They are mounted on ships to transfer cargo between ships and/or offshore structures and to conduct offshore construction. Boom cranes are also mounted on trucks for use in cargo transfer and construction sites. On the other hand, gantry and rotary cranes are used in fixed sites. Gantry cranes are widely used in the transportation industry, mines, steel mills, and assembly lines. Rotary cranes are mostly used in construction.

The high compliance of the cable–hook–payload assembly results in complex system dynamics. External (base) excitations at the suspension point can produce in-plane and out-of-plane pendulations as well as vertical oscillations of the payload. Even in the absence of external excitations, inertia forces due to the motion of the crane can induce significant payload pendulations. This problem is exacerbated by the fact that cranes are typically lightly damped, which means that any transient motion takes a long time to dampen out. Todd et al. (1997) report that the damping of ship-mounted boom cranes is 0.1–0.5% of their critical damping. Patel et al. (1987) offer a higher estimate of 1% for the vertical motion and 5% for the lateral motions. Willemstein et al. (1986), van den Boom et al. (1987, 1988), Patel et al. (1987), and Michelsen and Coppens (1988) found, using numerical simulation, that both stationary and transient dynamic forces due to payload motions are large enough that they need to be accounted for in the design and operation of cranes, thus emphasizing the need to predict and control both transient and stationary responses of the payload to excitations.

Suppression of payload pendulations/oscillations is especially important for offshore cranes. Wave-induced motions of the platform (a crane-ship or a semi-submersible) may contain significant energy near the natural frequency and/or twice the natural frequency of the free swinging load; this situation could initiate an external resonance and/or a parametric resonance. Therefore, the platform motions may induce large motions of the load directly or indirectly by creating a motion instability. For example, the platform motions may excite a parametric instability similar in form to that of the Mathieu instability (Nayfeh and Mook, 1979). This parametric instability has been observed at full scale by ship-crane operators and can arise in relatively mild sea states (McCormick and Witz, 1993).

Onshore cranes may also experience base excitations, leading to a complex dynamic response of the free swinging load, due to a variety of reasons, such as waves breaking on the shore and the interaction between the payload motion and the platform support system. However, this problem is most pronounced in offshore cranes. Assuming a work ability criterion based on the vertical displacement of the boom tip/payload only, Rawston and Blight (1978) calculated that a crane vessel in the North Sea could operate only for less than half of its availability time. Nojiri and Sasaki (1983) calculated that a barge crane in the East China Sea could only be used for heavy lifts for 34% of the time. More generally, payload pendulations/oscillations and the need to suppress them have been identified as a bottleneck in the operations of the transportation and construction industries even where relatively simple gantry cranes are concerned. Pendulation suppression is also necessary to increase the safety of operations and decrease the dynamic loads applied to the crane structure during operations (Brkić et al., 1998). Newly designed gantry cranes are larger, have higher lift capacities, and have greater lift heights and travel speeds, making the control of load pendulations a particular challenge (Champion, 1989).

The need for payload pendulation/oscillation suppression and the progress in computing facilities and sensors has led to mounting interest in crane control in recent years. However, most crane controllers developed up to now have been far from satisfactory. Once tested in actual operation, they were found to be cumbersome and ineffective and thus were left unused.

2. MODELING

Two approaches to the modeling of cranes are identified: *lumped-mass* and *distributed-mass* models.

2.1. Distributed-Mass Models

In this approach, the hoisting line is modeled as a distributed-mass cable and the hook and payload, lumped as a point mass, are applied as a boundary condition to this distributed-mass system. The only model available in this category is the planar model of d'Andrea-Novel et al. (1990, 1994) and d'Andrea-Novel and Boustany (1991b) for a gantry crane linearized around the cable's equilibrium position. They ignore the inertia of the payload and model the cable as a perfectly flexible, inextensible body using the wave equation

$$\rho \frac{\partial^2 w}{\partial t^2} - \frac{\partial}{\partial s}\left(\tau \frac{\partial w}{\partial s}\right) = 0 \tag{1}$$

where $w(s,t)$ is the transverse motion of the cable around its equilibrium position, s is a curvilinear coordinate representing the arclength along the cable, ρ is the mass per unit length of the cable, τ is the tension in the cable at equilibrium,

$$\tau(s) = mg + s\rho g \tag{2}$$

and m is the payload mass. The boundary conditions are

$$M\frac{\partial^2 w}{\partial t^2} - \tau\frac{\partial w}{\partial s} = F \qquad \text{at } s = 0 \tag{3}$$

$$\frac{\partial w}{\partial s} = 0 \qquad \text{at } s = \ell \tag{4}$$

where M is the mass of the trolley and F is the input force applied to the trolley. The boundary condition at the payload, equation (4), subjects the motion to the constraint

$$\frac{\partial^2 w}{\partial t^2} = 0 \qquad \text{at } s = \ell \tag{5}$$

thus ignoring the inertia of the payload. Joshi and Rahn (1995), Martindale et al. (1995), and Rahn et al. (1999) extended the model of d'Andrea-Novel and co-workers to include the inertia of the payload by changing the boundary condition at the payload to

$$m\frac{\partial^2 w}{\partial t^2} + \tau\frac{\partial w}{\partial s} = 0 \qquad \text{at } s = \ell \tag{6}$$

The model is valid only for a lumped mass m of the same order of magnitude as the mass of the cable and for small trolley displacement and cable angles. So it can only be used near the end of the trolley travel. However, even when a crane operates under no-load conditions, the mass of the hook is typically one order of magnitude heavier than the mass of the cable. As a result, this approach has limited practical applications.

2.2. Lumped-Mass Models

This is the most widely used approach to crane modeling. The hoisting line is modeled as a massless cable. The payload is lumped with the hook and modeled as a point mass. The cable–hook–payload assembly is modeled as a spherical pendulum. The resulting mathematical representation is simple and compact while capturing the complex dynamics of the payload motion.

There are two classes of lumped-mass model, depending on the way the external excitations are introduced to the system, namely *reduced* and *extended* models. A reduced model lumps all external excitations into expressions representing the motion of the pendulum suspension point (base excitations). This approach assumes that the payload motions are influenced by, but do not have a significant influence on, the platform motion; that is, the inertial coordinates ξ, ζ, and η of the suspension point (as shown in Figure 4) are known functions of time t. An extended model adds the crane support mechanism and the platform to the dynamic model, thereby incorporating the interactions among the support mechanism, the platform, and the cable–payload assembly in the model.

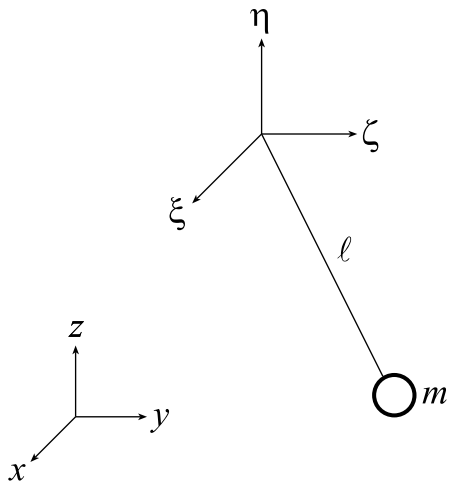


Figure 4. Schematic and coordinate system of the lumped-mass model.

All reduced models are special cases of the same classical model of a spherical pendulum under base excitations. On the other hand, each extended model is a unique system capturing a distinct set of the crane-structure dynamics. In the following, we analyze the reduced crane model and then discuss the extended crane models available in the literature.

3. THE REDUCED MODEL

We consider a pendulum of length ℓ and mass m and express the Cartesian coordinates of the suspension point and mass as $[\xi(t), \zeta(t), \eta(t)]$ and $[x(t), y(t), z(t)]$, respectively. These coordinates have to satisfy the constraint

$$(x-\xi)^2 + (y-\zeta)^2 + (z-\eta)^2 = (\ell + r)^2 \tag{7}$$

where r is the elastic stretch in the cable. The Lagrangian of the system is

$$\mathcal{L} = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz - \tfrac{1}{2}c_e r^2 \tag{8}$$

where c_e is the longitudinal stiffness of the cable.

To determine the equations of motion, we use equation (7) to substitute for z and $\dot{z}$ into the Lagrangian, apply the Euler–Lagrange equations, and obtain (Chin et al., 1998)

$$\ddot{x} = -\frac{x-\xi}{\sqrt{\lambda}}(g + \ddot{\eta}) - \frac{x-\xi}{4\lambda^2}\dot{\lambda}^2 + \frac{x-\xi}{2\lambda}\ddot{\lambda} \tag{9}$$

$$\ddot{y} = -\frac{y-\zeta}{\sqrt{\lambda}}(g + \ddot{\eta}) - \frac{y-\zeta}{4\lambda^2}\dot{\lambda}^2 + \frac{y-\zeta}{2\lambda}\ddot{\lambda} \tag{10}$$

$$\ddot{r} = -\frac{c\,r}{m(\ell+r)^2}\lambda + \frac{g+\ddot{\eta}}{\ell+r}\sqrt{\lambda} + \frac{\dot{\lambda}^2}{4\lambda\,(\ell+r)}$$
$$-\frac{1}{\ell+r}\Big((\dot{\ell}+\dot{r})^2 - (\dot{y}-\dot{\zeta})^2 - (\dot{x}-\dot{\xi})^2 + (\ell+r)\ddot{\ell}$$
$$-(y-\zeta)(\ddot{y}-\ddot{\zeta}) - (x-\xi)(\ddot{x}-\ddot{\xi})\Big) \tag{11}$$

where $\lambda = (\ell+r)^2 - (x-\xi)^2 - (y-\zeta)^2$. These are the *exact* equations of motion of the spherical pendulum model in its most generic form.

To apply perturbation analysis to this system, Chin et al. (1998) wrote the cable length as $\ell = \ell_o + \ell_c$ and the cable stretch as $r = r_o + r$, where ℓ_o is the cable length at some reference configuration, ℓ_c is the change in the cable length, and r_o and r are, respectively, the static and dynamic stretches in the cable. Assuming x, y, r to be of the same order of magnitude and considering a slow variation of the cable length $\ell_c \leq O(x^2)$, Chin et al. (1998) extracted a third-order approximation of equations (9)–(11), used it to model a ship-mounted boom crane, and obtained

$$\ddot{x} + \omega_1^2 x = -\ddot{\xi} + \frac{x}{\ell_{cr}}\left(\ddot{\ell}_c - \ddot{\eta}\right) + \frac{\omega_1^2}{\ell_{cr}}(\ell_c x + r(x-\xi))$$
$$-\omega_1^2\frac{x}{\ell_{cr}^2}\left(r^2 + \tfrac{1}{2}x^2 + \tfrac{1}{2}y^2\right) - \frac{x}{\ell_{cr}^2}\left(\dot{x}^2 + \dot{y}^2\right)$$
$$-\frac{x}{\ell_{cr}^2}(x\ddot{x} + y\ddot{y} + r\ddot{r}) + \frac{\ddot{r}}{\ell_{cr}}(x-\xi) \tag{12}$$

$$\ddot{y} + \omega_1^2 y = -\ddot{\zeta} + \frac{y}{\ell_{cr}}\left(\ddot{\ell}_c - \ddot{\eta}\right) + \frac{\omega_1^2}{\ell_{cr}}(\ell_c y + r(y-\zeta))$$
$$-\omega_1^2\frac{y}{\ell_{cr}^2}\left(r^2 + \tfrac{1}{2}x^2 + \tfrac{1}{2}y^2\right) - \frac{y}{\ell_{cr}^2}\left(\dot{x}^2 + \dot{y}^2\right)$$
$$-\frac{y}{\ell_{cr}^2}(x\ddot{x} + y\ddot{y} + r\ddot{r}) + \frac{\ddot{r}}{\ell_{cr}}(y-\zeta) \tag{13}$$

$$\ddot{r} + \omega_2^2 r = \ddot{\eta} - \ddot{\ell}_c + \frac{\omega_1^2}{\ell_{cr}}(\tfrac{1}{2}x^2 - x\xi + \tfrac{1}{2}y^2 - y\zeta) - \frac{2}{\ell_{cr}}(\dot{x}\dot{\xi} + \dot{y}\dot{\zeta})$$
$$+\omega_1^2\frac{r}{\ell_{cr}^2}(r^2 + \tfrac{1}{2}x^2 + \tfrac{1}{2}y^2) + \frac{(\ell_{cr}-r)}{\ell_{cr}^2}(\dot{x}^2 + \dot{y}^2) - \frac{2\dot{r}}{\ell_{cr}^2}(x\dot{x} + y\dot{y})$$
$$+\frac{(\ell_{cr}-r)}{\ell_{cr}^2}(x\ddot{x} + y\ddot{y}) - \frac{\ddot{r}}{\ell_{cr}^2}(x^2 + y^2) - \frac{1}{\ell_{cr}}(\xi\ddot{x} + x\ddot{\xi} + \zeta\ddot{y} + y\ddot{\zeta}) \tag{14}$$

where $\ell_{cr} = \ell_o + r_o$ is the characteristic cable length, $\omega_1^2 = g/\ell_{cr}$ is the natural frequency of payload pendulations, and $\omega_2^2 = c_e/m$ is the natural frequency of the longitudinal oscillations.

A special case of interest allows for the reeling and unreeling of an inextensible cable. The equations of motion of the payload can, thus, be reduced to two equations

$$\ddot{x} + \omega^2 x = -\ddot{\xi} + \frac{x}{\ell_o}\left(\ddot{\ell}_c - \ddot{\eta}\right) - \frac{x}{\ell_o^2}\left(\tfrac{\omega^2}{2}\left(x^2+y^2\right) + \left(\dot{x}^2+\dot{y}^2\right) + \left(x\ddot{x}+y\ddot{y}\right)\right) \tag{15}$$

$$\ddot{y} + \omega^2 y = -\ddot{\zeta} + \frac{y}{\ell_o}\left(\ddot{\ell}_c - \ddot{\eta}\right) - \frac{y}{\ell_o^2}\left(\tfrac{\omega^2}{2}\left(x^2+y^2\right) + \left(\dot{x}^2+\dot{y}^2\right) + \left(x\ddot{x}+y\ddot{y}\right)\right) \tag{16}$$

where $\omega^2 = g/\left(\ell_o + \ell_c\right)$. For the same assumptions and using a spherical *body* coordinate system attached to the suspension point, Figure 5, the *exact* equations of motion can be written as

$$\begin{aligned}
\ddot{\theta}\,\cos\phi + \omega^2 \sin\theta \;=\; & 2\,\dot{\theta}\,\dot{\phi}\,\sin\phi + 2\dot{\beta}\left(\dot{\phi}\,\sin\phi + \frac{\dot{\xi}}{\ell}\sin\theta + \frac{\dot{\eta}}{\ell}\cos\theta\right) \\
& -2\,\dot{\gamma}\,\cos\theta\left(\dot{\phi}\,\cos\phi + \frac{\dot{\zeta}}{\ell}\right) \\
& -2\,\frac{\dot{\ell}}{\ell}\left(\dot{\theta}\,\cos\phi + \dot{\beta}\,\cos\phi + \dot{\gamma}\,\cos\theta\,\sin\phi\right) \\
& -\dot{\beta}^{\,2}\left(\frac{\xi}{\ell}\cos\theta - \frac{\eta}{\ell}\sin\theta\right) - \dot{\gamma}^{\,2}\left(\frac{\xi}{\ell}\cos\theta - \tfrac{1}{2}\sin 2\theta\,\cos\phi\right) \\
& -\dot{\beta}\,\dot{\gamma}\,\sin\theta\left(\frac{\zeta}{\ell} + \sin\phi\right) - \ddot{\beta}\left(\cos\phi - \frac{\xi}{\ell}\sin\theta - \frac{\eta}{\ell}\cos\theta\right) \\
& -\ddot{\gamma}\,\cos\theta\left(\sin\phi + \frac{\zeta}{\ell}\right) + \frac{\ddot{\xi}}{\ell}\cos\theta - \frac{\ddot{\eta}}{\ell}\sin\theta
\end{aligned} \tag{17}$$

$$\begin{aligned}
\ddot{\phi} + \omega^2\cos\theta\sin\phi \;=\; & -\tfrac{1}{2}\dot{\theta}^{\,2}\sin 2\phi - \dot{\beta}\left(\dot{\theta}\sin 2\phi - 2\,\frac{\dot{\xi}}{\ell}\cos\theta\,\sin\phi + 2\,\frac{\dot{\eta}}{\ell}\sin\theta\,\sin\phi\right) \\
& +2\,\dot{\gamma}\left(\dot{\theta}\cos\theta\cos^2\phi - \frac{\dot{\xi}}{\ell}\cos\phi + \frac{\dot{\zeta}}{\ell}\sin\theta\,\sin\phi\right) - 2\,\frac{\dot{\ell}}{\ell}\left(\dot{\phi} - \dot{\gamma}\sin\theta\right) \\
& -\dot{\beta}^{\,2}\left(\tfrac{1}{2}\sin 2\phi - \frac{\xi}{\ell}\sin\theta\,\sin\phi - \frac{\eta}{\ell}\cos\theta\,\sin\phi\right) \\
& +\dot{\gamma}^{\,2}\left(\tfrac{1}{2}\cos^2\theta\,\sin 2\phi + \frac{\xi}{\ell}\sin\theta\,\sin\phi + \frac{\zeta}{\ell}\cos\phi\right) \\
& +\dot{\beta}\,\dot{\gamma}\left(\cos\theta\,\cos 2\phi - \frac{\zeta}{\ell}\cos\theta\,\sin\phi - \frac{\eta}{\ell}\cos\phi\right) \\
& +\ddot{\beta}\,\sin\phi\left(\frac{\xi}{\ell}\cos\theta - \frac{\eta}{\ell}\sin\theta\right) \\
& +\ddot{\gamma}\left(\sin\theta - \frac{\xi}{\ell}\cos\phi + \frac{\zeta}{\ell}\sin\theta\,\sin\phi\right) \\
& -\frac{\ddot{\xi}}{\ell}\sin\theta\,\sin\phi - \frac{\ddot{\zeta}}{\ell}\cos\phi - \frac{\ddot{\eta}}{\ell}\cos\theta\,\sin\phi
\end{aligned} \tag{18}$$

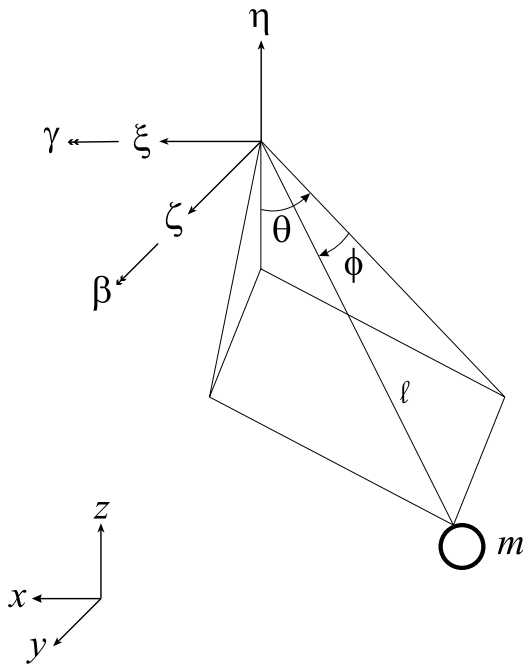


Figure 5. Spherical coordinate system.

where $[0, \beta, \gamma]$ is the rotation vector of the suspension point (β is the crane luff angle and γ is the crane slew angle), θ is the in-plane angle, and ϕ is the out-of-plane angle between the equilibrium position of the cable–payload assembly and the cable at time t.

The standard model of a spherical pendulum, however, assumes both an inextensible cable and a constant length cable. In the following, we develop and analyze this model in detail. To determine the equations of motion to third order in x and y, we let ξ, ζ, and η be $\leq O(x, y)$. It follows from equation (7) that

$$z = \eta - \ell + \frac{(x-\xi)^2 + (y-\zeta)^2}{2\ell} + \frac{[(x-\xi)^2 + (y-\zeta)^2]^2}{8\ell^3} + \cdots. \tag{19}$$

Substituting equation (19) into equation (8), keeping up to quartic terms, and letting $\omega^2 = g/\ell$, we obtain

$$\begin{aligned}\mathcal{L} &= \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{\eta}^2) + \frac{m}{\ell}\dot{\eta}\left[(x-\xi)(\dot{x}-\dot{\xi}) + (y-\zeta)(\dot{y}-\dot{\zeta})\right] \\ &\quad + \frac{m}{2\ell^2}\left[(x-\xi)(\dot{x}-\dot{\xi}) + (y-\zeta)(\dot{y}-\dot{\zeta})\right]^2 - mg(\eta - \ell) \\ &\quad - \tfrac{1}{2}m\omega^2[(x-\xi)^2 + (y-\zeta)^2] - \frac{m\omega^2}{8\ell^2}[(x-\xi)^2 + (y-\zeta)^2]^2 + \cdots. \end{aligned} \tag{20}$$

Applying the Euler–Lagrange equations and adding linear damping ordered at $\mu \leq O(x^2)$ yields

$$\begin{aligned}\ddot{x} + 2\mu\dot{x} + \omega^2 x &= \omega^2\xi - \frac{(x-\xi)}{\ell}\ddot{\eta} - \omega^2\frac{(x-\xi)}{2\ell^2}\left[(x-\xi)^2 + (y-\zeta)^2\right] \\ &\quad - \frac{(x-\xi)}{\ell^2}\left[(\dot{x}-\dot{\xi})^2 + (\dot{y}-\dot{\zeta})^2 + (x-\xi)(\ddot{x}-\ddot{\xi})\right. \\ &\quad \left. + (y-\zeta)(\ddot{y}-\ddot{\zeta})\right] \qquad (21)\end{aligned}$$

$$\begin{aligned}\ddot{y} + 2\mu\dot{y} + \omega^2 y &= \omega^2\zeta - \frac{(y-\zeta)}{\ell}\ddot{\eta} - \omega^2\frac{(y-\zeta)}{2\ell^2}\left[(x-\xi)^2 + (y-\zeta)^2\right] \\ &\quad - \frac{(y-\zeta)}{\ell^2}\left[(\dot{x}-\dot{\xi})^2 + (\dot{y}-\dot{\zeta})^2 + (x-\xi)(\ddot{x}-\ddot{\xi})\right. \\ &\quad \left. + (y-\zeta)(\ddot{y}-\ddot{\zeta})\right] \qquad (22)\end{aligned}$$

where the damping μ is assumed to be symmetric in both pendulation directions. The equations are symmetric in the pairs (x, ξ) and (y, ζ), reflecting the physical symmetry of the in-plane and out-of-plane motions of the payload. As a result, the linear natural frequencies ω of the payload pendulations are identical. Furthermore, the equations show that the in-plane and out-of-plane modes are coupled by cubic terms, representing the geometric and kinetic nonlinearities in the model.

3.1. Approximate Solution of the Reduced Model

The existence of cubic nonlinearities and the symmetry between the dynamics of the in-plane and out-plane directions produce a one-to-one internal (autoparametric) resonance, leading to complex dynamics and energy exchange between the two modes (Nayfeh, 2000). Experiments by Todd et al. (1997) have shown that a ship-mounted boom crane exhibits this predicted dynamic behavior under external forcing, including chaotic and/or nonplanar responses to strictly planar excitations at frequencies near the natural frequency of pendulations.

The lateral ξ and ζ and vertical η motions of the boom tip produce external (additive) as well as parametric (multiplicative) excitations, respectively. To examine the response of the system to a general forcing near the resonance frequencies, we order the external excitations ξ and ζ at $\leq O(x^3)$ and the parametric excitation η at $\leq O(x^2)$. Expanding equations (21) and (22) and dropping terms of order higher than $O(x^3)$, we obtain

$$\ddot{x} + 2\mu\dot{x} + \omega^2 x + \frac{\omega^2}{2\ell^2}x\left(x^2+y^2\right) + \frac{x}{\ell^2}\left(\dot{x}^2 + \dot{y}^2 + x\ddot{x} + y\ddot{y}\right) = -\ddot{\xi} - \frac{x}{\ell}\ddot{\eta} \qquad (23)$$

$$\ddot{y} + 2\mu\dot{y} + \omega^2 y + \frac{\omega^2}{2\ell^2}y\left(x^2+y^2\right) + \frac{y}{\ell^2}\left(\dot{x}^2 + \dot{y}^2 + x\ddot{x} + y\ddot{y}\right) = -\ddot{\zeta} - \frac{y}{\ell}\ddot{\eta}\,. \qquad (24)$$

We use the method of multiple scales (Nayfeh, 1973, 1981) to determine a first-order approximate solution of equations (23) and (24) for small- but finite-amplitude motions. The worst-case excitation of the crane is a combination of direct excitations at the natural

frequency (primary resonance) and parametric excitation at twice the natural frequency (principal parametric resonance). To this end, we introduce a small dimensionless parameter ϵ as a bookkeeping device and the time scales

$$T_0 = t \quad \text{and} \quad T_2 = \epsilon^2 t. \tag{25}$$

In terms of these scales, the time derivatives become

$$\frac{\mathrm{d}}{\mathrm{d}t} = D_0 + \epsilon^2 D_2 + \cdots \tag{26}$$

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} = D_0^2 + 2\epsilon^2 D_0 D_2 + \cdots \tag{27}$$

where $D_n \equiv \partial/\partial T_n$. We apply the displacement combination

$$\xi = \epsilon^3 u_0 \cos\Omega_1 t, \quad \zeta = \epsilon^3 v_0 \cos\Omega_1 t, \quad \text{and} \quad \eta = \epsilon^2 w_0 \cos\Omega_2 t \tag{28}$$

with a primary excitation frequency

$$\Omega_1 = \omega + \epsilon^2 \sigma_1 \tag{29}$$

and a principal parametric excitation frequency

$$\Omega_2 = 2\omega + \epsilon^2 \sigma_2 \tag{30}$$

where σ_1 and σ_2 are detuning parameters. We seek a uniform approximate solution of equations (23) and (24) in the form

$$x(t;\epsilon) \simeq \epsilon x_1(T_0, T_2) + \epsilon^3 x_2(T_0, T_2) \tag{31}$$

$$y(t;\epsilon) \simeq \epsilon y_1(T_0, T_2) + \epsilon^3 y_2(T_0, T_2). \tag{32}$$

Substituting equations (25)–(32) into equations (23) and (24) and equating coefficients of like powers of ϵ leads to the following problems:
Order ϵ

$$D_0^2 x_1 + \omega^2 x_1 = 0 \tag{33}$$

$$D_0^2 y_1 + \omega^2 y_1 = 0; \tag{34}$$

Order ϵ^3

$$\begin{aligned} D_0^2 x_2 + \omega^2 x_2 &= -2D_0 D_2 x_1 - 2\mu D_0 x_0 + \omega^2 u_0 \cos\Omega_1 T_0 - \frac{4\omega^2}{\ell} w_0 x_1 \cos\Omega_2 T_0 \\ &\quad - x_1 (D_0 x_1 D_0 x_1 + x_1 D_0^2 x_1 + D_0 y_1 D_0 y_1 + y_1 D_0^2 y_1) \end{aligned} \tag{35}$$

$$\begin{aligned} D_0^2 y_2 + \omega^2 y_2 &= -2D_0 D_2 y_1 - 2\mu D_0 y_0 + \omega^2 v_0 \cos\Omega_1 T_0 - \frac{4\omega^2}{\ell} w_0 y_1 \cos\Omega_2 T_0 \\ &\quad - y_1 (D_0 x_1 D_0 x_1 + x_1 D_0^2 x_1 + D_0 y_1 D_0 y_1 + y_1 D_0^2 y_1). \end{aligned} \tag{36}$$

The solutions of equations (33) and (34) can be expressed as

$$x_1 = A_1(T_2)\mathrm{e}^{\mathrm{i}\omega T_0} + \mathrm{cc} \quad \text{and} \quad y_1 = A_2(T_2)\mathrm{e}^{\mathrm{i}\omega T_0} + \mathrm{cc} \tag{37}$$

where cc indicates the complex conjugate of the preceding terms. Substituting equations (37) into equations (35) and (36) and eliminating the terms that produce secular terms, we obtain the following modulation equations

$$A_1' = -\tfrac{1}{4}\mathrm{i}\omega\bar{A}_1(A_1^2 + 3A_2^2) + \tfrac{1}{2}\mathrm{i}\omega A_1 A_2\bar{A}_2 - \mu A_1 - \frac{u_0}{4}\mathrm{i}\omega\mathrm{e}^{\mathrm{i}\sigma_1 T_2} + \frac{w_0}{\ell}\mathrm{i}\omega\bar{A}_1\mathrm{e}^{\mathrm{i}\sigma_2 T_2} \tag{38}$$

$$A_2' = -\tfrac{1}{4}\mathrm{i}\omega\bar{A}_2(A_2^2 + 3A_1^2) + \tfrac{1}{2}\mathrm{i}\omega A_1\bar{A}_1 A_2 - \mu A_2 - \frac{v_0}{4}\mathrm{i}\omega\mathrm{e}^{\mathrm{i}\sigma_1 T_2} + \frac{w_0}{\ell}\mathrm{i}\omega\bar{A}_2\mathrm{e}^{\mathrm{i}\sigma_2 T_2} \tag{39}$$

where the prime indicates the derivative with respect to the slow time scale T_2.

To determine the slow variations of the amplitudes and phases of the solution, we introduce the polar transformation

$$A_k(T_2) = \tfrac{1}{2}a_k(T_2)\mathrm{e}^{\mathrm{i}\beta(T_2)}, \quad k = 1, 2 \tag{40}$$

into equations (38) and (39) and obtain

$$\begin{aligned} a_1' &= -\mu a_1 - \tfrac{1}{2}u_0\omega\sin(\beta_1 - \sigma_1 T_2) + \frac{w_0}{\ell}\omega\, a_1\sin(2\beta_1 - \sigma_2 T_2) \\ &\quad - \tfrac{3}{16}\omega\, a_1 a_2^2\sin(2\beta_1 - 2\beta_2) \end{aligned} \tag{41}$$

$$\begin{aligned} \beta_1' &= -\frac{u_0}{2\,a_1}\,\omega\cos(\beta_1 - \sigma_1 T_2) + \frac{w_0}{\ell}\,\omega\cos(2\beta_1 - \sigma_2 T_2) \\ &\quad - \tfrac{1}{16}\omega\, a_1^2 + \tfrac{1}{8}\omega\, a_2^2 - \tfrac{3}{16}\,\omega\, a_2^2\cos(2\beta_1 - 2\beta_2) \end{aligned} \tag{42}$$

$$\begin{aligned} a_2' &= -\mu a_2 - \tfrac{1}{2}\,v_0\,\omega\,\sin(\beta_2 - \sigma_1 T_2) + \frac{w_0}{\ell}\,\omega\, a_2\sin(2\beta_2 - \sigma_2 T_2) \\ &\quad + \tfrac{3}{16}\,\omega\, a_1^2\, a_2\sin(2\beta_1 - 2\beta_2) \end{aligned} \tag{43}$$

$$\begin{aligned} \beta_2' &= -\frac{v_0}{2\,a_2}\,\omega\cos(\beta_2 - \sigma_1 T_2) + \frac{w_0}{\ell}\,\omega\,\cos(2\beta_2 - \sigma_2 T_2) \\ &\quad + \tfrac{1}{8}\,\omega\, a_1^2 - \tfrac{1}{16}\omega\, a_2^2 - \tfrac{3}{16}\,\omega\, a_1^2\cos(2\beta_1 - 2\beta_2). \end{aligned} \tag{44}$$

The cubic terms in equations (41) and (43) indicate the possibility of an exchange of energy between the two modes. The presence of the amplitudes a_1 and a_2 in the denominator in equations (42) and (44) leads to instabilities in the numerical integration of the modulation equations whenever either amplitude approaches zero. Therefore, a Cartesian transformation instead of this polar transformation is usually used to write the modulation equations (Nayfeh, 2000). While it is harder to discern the system's behavior by inspection of those equations, they do not pose any difficulties to numerical integration.

Miles (1962, 1984) used this model to examine the response of a lightly damped, spherical pendulum to a simple harmonic, planar displacement of the suspension point. He found that nonplanar motions could be excited due to the nonlinear interaction between the two modes. Chin and Nayfeh (1996) and Chin et al. (2001) used the model to study ship-mounted crane dynamics in two cases of harmonic base excitations at the boom tip: the

case of primary resonance and the case of principal parametric resonance. They found that, while the parametric excitation exhibits principal parametric resonance in the neighborhood of twice the natural frequency of the system, the response is always periodic and planar. On the other hand, direct excitations produce complex dynamics when the excitation frequency approaches the natural frequency of the system (primary resonance). They also found that a strictly planar excitation could produce in-plane and out-of-plane pendulations and that the response may exhibit sudden jumps, modulation of the response amplitudes and phases (quasi-periodic or two-torus motion), and chaos.

Using the method of multiple scales, Chin et al. (1998) solved equations (12)–(14) analytically and numerically. They found out that a parametric excitation at twice the natural frequency leads to a sudden jump in the response as the cable is unreeled. They also demonstrated that introducing a harmonic change in the cable length at the same frequency as the excitation can suppress this dynamic instability and result in a smooth response.

Abdel-Rahman and Nayfeh (2000) used the variable cable length model, equations (15) and (16), to study a boom crane allowing for reeling and unreeling of an inextensible cable at a constant speed. The analytical solution and numerical simulation show that a planar direct base excitation near the natural frequency can produce in-plane and out-of-plane motions, sudden jumps in the response, and a chaotic response as the cable length is changed.

Elling and McClinton (1973) were the first to examine the nonlinearities involved in the dynamic response of a boom crane. They modeled the crane as a spherical pendulum undergoing a conical motion (Greenwood, 1988), while the hoisting cable length is changed at a constant speed. As a result, the model assumed the motion in one direction to be of the order of $O(x)$ and in the other direction to be of the order of $O(x^3)$ and neglected the potential energy in the higher-order direction. In practice, the payload pendulates around the suspension point rather than following a prescribed circular path. Furthermore, whenever a nonplanar motion occurs, the motions in both directions have the same order of magnitude and the potential energy in both directions cannot be ignored. Using numerical simulation, they solved the equations of motion subject to harmonic base excitations and found a resonant response when the excitation frequency is near the natural frequency (primary resonance) or one-half the natural frequency (secondary resonance) of the assembly. They also concluded that unreeling of the cable dampens the payload motion while reeling of the cable excites it further.

3.2. Reduced Linear Model

Two special cases of the classical model are of particular interest: the linear three-dimensional model and the nonlinear planar model. Assuming small motions, we can linearize the generalized model around the payload equilibrium point. Dropping higher-order terms from equations (23) and (24) results in the reduced linear model:

$$\ddot{x} + 2\mu\,\dot{x} + \omega^2 x = F_x \tag{45}$$

$$\ddot{y} + 2\mu\,\dot{y} + \omega^2 y = F_y\,. \tag{46}$$

This is the widely used linear crane model. While it is quite simple, thus amenable, to use as a model of the plant in controller designs, it neglects the nonlinear terms, thus masking the dynamic complexities arising from coupling of the two modes. Also, this approximation is

only valid as long as the payload motions remain small. Jones and Petterson (1988) report that, when pendulations reach a maximum swing amplitude greater than a few degrees, the nonlinear aspects of the swinging object must be taken into account. So the linear model breaks down whenever in-plane pendulations grow or out-of-plane pendulations occur. Furthermore, dropping the cubic nonlinearities on the left-hand sides of equations (23) and (24), while retaining the direct excitations on the right-hand sides, even though both are of the order of $O(x^3)$, is inconsistent. On the other hand, introducing hard excitations of the order of $O(x)$ will produce large motions, thus requiring the inclusion of the cubic nonlinearities on the left-hand side.

3.3. Reduced Planar Model

Setting the out-of-plane motion y equal to zero in equation (23), we obtain the nonlinear two-dimensional model of the pendulum up to third order $O(x^3)$ as

$$\ddot{x} + 2\mu\,\dot{x} + \omega^2 x + \frac{\omega^2}{2\ell^2}x^3 + \frac{x}{\ell^2}\left(\dot{x}^2 + x\ddot{x}\right) = -\,\ddot{\xi} \;-\frac{x}{\ell}\ddot{\eta}\,. \tag{47}$$

Similarly, setting the out-of-plane angle ϕ and the slew angle γ equal to zero in equation (17) and assuming a constant cable length, we obtain the *exact* nonlinear two-dimensional model of the pendulum in a more compact form

$$\begin{aligned}\ddot{\theta} \;+ 2\mu\,\dot{\theta} \;+ \omega^2 \sin\theta \;&= 2\dot{\beta}\left(\frac{\dot{\xi}}{\ell}\sin\theta \;+\frac{\dot{\eta}}{\ell}\cos\theta\right) - \dot{\beta}^{\,2}\left(\frac{\xi}{\ell}\cos\theta \;-\frac{\eta}{\ell}\sin\theta\right)\\ &\quad + \ddot{\beta}\left(\frac{\xi}{\ell}\sin\theta \;+\frac{\eta}{\ell}\cos\theta\right) + \frac{\ddot{\xi}}{\ell}\cos\theta \;-\frac{\ddot{\eta}}{\ell}\sin\theta\,.\end{aligned} \tag{48}$$

This is the classical planar pendulum equation of motion. The higher-order terms involving β, the luff angle, represent the dynamic effect of the rotation of the body coordinate system attached to the suspension point. While a first-order approximation of the equation may neglect all of these terms, that would render the equation approximate and valid only at low luffing speeds.

4. EXTENDED MODELS

4.1. Gantry Cranes

The most popular extended model of gantry cranes, Figure 6, augments the planar version of the pendulum model, equation (48), with the trolley's planar equation of motion. The result is

$$M\ddot{\xi} \;= F - T\sin\theta \;- f \tag{49}$$

where

$$T \;=\; mg\cos\theta \;+ m\ell\dot{\theta}^{\,2} - m\ddot{\xi}\;\sin\theta \tag{50}$$

$$f \;=\; \mu_t\,(Mg + T\cos\theta\,) \tag{51}$$

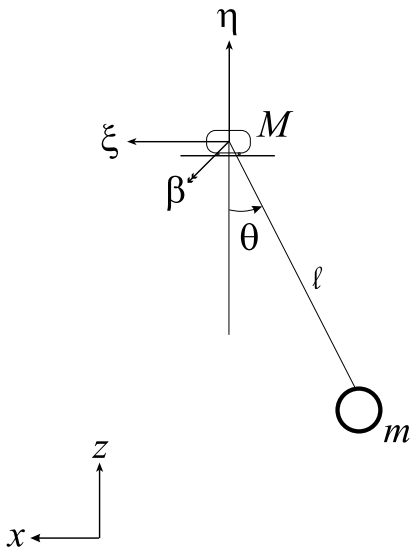


Figure 6. Schematic diagram of an extended gantry crane model.

T is the tension in the cable, f is the friction force between the trolley wheels and the girder, and μ_t is the friction coefficient. It should be noted that, under this configuration, the linear natural frequency of the pendulum is dependent on the trolley and payload masses; that is,

$$\omega_p = \sqrt{\frac{(M+m)g}{M\ell}}. \tag{52}$$

Field (1961) further augmented this model with an equation describing the dynamics of the cable hoisting motion. Auernig and Troger (1987) *partially* linearized the model with respect to the pendulation angle θ .

Moustafa and Ebeid (1988) derived the only nonlinear model of a bidirectional (double girder) gantry crane available in the literature. The model accounts for rigid-body motions of the payload, translation of the trolley and the moving girder, the torsional stiffness and a constant hoisting speed of the cable, and the inertia of the trolley and girder driving motors. They also presented a version of the model linearized around the equilibrium position of the payload. Ebeid et al. (1992) augmented the linearized model with a linear model of the trolley and girder motors accounting for the motor dynamics. Using numerical simulation, they found that the motors introduce linear damping into the system response.

Zrnić et al. (1997, 1998) presented a linear model of a bidirectional gantry crane by lumping the bridge mass into five point masses, each of the bridge legs into a point mass, and the payload as another point mass. The model accounts for linear structural stiffness and damping in the bridge and legs and for the linear stiffness and damping in the driving motor of each leg.

4.2. Rotary Cranes

Parker et al. (1995a) linearized the spherical pendulum model, equations (23) and (24), and applied a base excitation representing the slew motion (jib rotation) of a rotary crane, but did not apply the trolley motion along the jib. Parker et al. (1995b) extended this model to account for a reeling of the cable synchronized with the slew motion. The resulting linear time-varying model is inconsistent. It neglects the cubic nonlinearities representing geometric and kinetic nonlinearities in the model, while retaining first-order direct and second-order parametric base excitations, which are equivalent to cubic and quartic nonlinear motion terms. Parker et al. (1996) extended the model of Parker et al. (1995b) to account for trolley translation along the jib and dropped the higher-order base excitations. However, the linearized model still contained direct excitations of the order of $O(x^3, y^3)$.

Brkić et al. (1998) presented a linear, planar model of a rotary crane by lumping the jib and tower mass into eight point masses and each of the hoisting mechanism, trolley, and payload as a point mass. The model accounts for linear structural stiffness in the jib and tower and for the linear stiffness hoisting motor and cable, while neglecting the damping in the system.

4.3. Boom Cranes

Ito et al. (1978) studied a truck-mounted crane by modeling the cable–hook–payload assembly as a spherical pendulum, the boom as a beam, and the hoisting and power lines as linearly viscoelastic bodies subject to engine torque and friction in the bearings. They carried out numerical simulations on a special planar case where all accelerations are set equal to zero; they also carried out experiments for this case. The results show that the payload undergoes oscillations due to longitudinal vibrations in the boom luffing line and the hoisting cable and pendulum-like payload oscillations. The period of this oscillation is an order of magnitude larger than the periods of the longitudinal vibrations of the lines.

Sakawa and Nakazumi (1985) augmented the spherical pendulum with two equations representing the rotational motions of the crane base (slew) and the boom (luff). Sato and Sakawa (1988) extended this model to include an extra jib at the boom tip with a flexible joint. Both models were partially linearized with respect to the in-plane pendulation angle. Souissi and Koivo (1992) extended the model of Sakawa and Nakazumi (1985) by coupling the system equations with an equation describing the reeling of the hoisting line.

Patel et al. (1987), McCormick and Witz (1993), and Witz (1995) used a linear model of a ship-mounted crane to study the coupled motions of the ship and crane. They modeled the ship as a rigid body, the boom as a beam under bending, the hoisting and mooring lines as elastic bodies, and the payload as a point mass undergoing spherical pendulum-like oscillations. Added mass and damping, due to ship–sea interaction, were also included in the model. Using computer simulations, they solved a planar version of the model, including the surge, heave, and pitch of the ship and the surge and heave of the payload for ship and payload motions in head seas. They found that "resultant vessel heave motions ... are not significantly affected by the vertical dynamics" and that "pitch motions ... are only slightly affected by vertical dynamics of the hook load." In beam seas, they found out that "the influence of coupling on sway motions of the vessel is very small," and that "the hook load does not significantly affect the roll motions of the vessel." They concluded that coupling of ship motions with payload motions has negligible influence on ship motions away from the

natural frequency of the crane. The calculations of Nojiri and Sasaki (1983) have shown that payload pendulations near the resonance frequency of the cable–payload assembly have a pronounced effect on the roll and pitch motions of a crane vessel in both regular and irregular waves. Furthermore, Patel et al. (1987), McCormick and Witz (1993), and Witz (1995) reduced a planar model of the crane to a Mathieu equation, thus showing that the load can be parametrically excited due to the relative motion between the boom tip and load. Based on the stability diagrams of the Mathieu equation, they derived the operability conditions for the crane.

Schellin et al. (1989) augmented the linear planar pendulum model, equation (45), with an equation describing the stretch motion in the hoisting line and three equations of motion describing the planar rigid-body motion of the ship (surge, heave, and pitch). The ship and boom were modeled as a rigid body, the mooring system as a nonlinear restoring force, and the hoisting cable was allowed to stretch elastically. Added mass and damping due to the ship–sea interaction were also included in the model. They found that the natural frequency of the cable–payload assembly in stretching is four orders of magnitude higher than that in pendulation. They also reported strong coupling between the payload pendulation and the ship surge and pitch motions when the crane is excited near its natural frequency. Numerical simulations of the system revealed chaos in the response of the load to regular waves at a wave frequency near the natural frequency of "the hook load" and period doubling at frequencies further afield from it.

Schellin et al. (1991) extended this model to three dimensions by coupling the spherical pendulum model to a linear model of the ship rigid-body motions and an equation describing the cable stretch as a viscoelastic body. They compared the simulation results of the model and those of a linearized version of the model to the results of experiments on a ship model and found that the nonlinear model was "more realistic" than the linearized model near resonance where large ship and load motions occur. Simulations of the response to wave groups show that, when the frequencies of the component waves of the wave group are close to the natural frequency of the cable-payload assembly, the response is chaotic. They also found that payload motions induce ship yaw; however "horizontal ship motions did not noticeably affect load oscillations." There is a need for more elaborate models to further examine the coupling between the motions of the crane and those of the ship before this coupling can be neglected or factored into crane models.

Posiadała et al. (1990) modeled the cable–payload assembly in a truck-mounted crane as a spherical pendulum. Base excitations due to the boom slew, luff, and telescopic (extension) motion and forcing due to cable reeling/unreeling were introduced into the equations of motion. They used numerical simulations to calculate the forced payload response under various motion combinations for 10 s and then its free response for the subsequent 10 s. They found that, except in the absence of slew motions, the payload response is three-dimensional and cannot be considered as a planar phenomenon. Posiadała et al. (1991) extended the model to account for the flexibility of the cable being modeled as a Kelvin–Voigt body. The pendulum equations of motion were augmented with an equation describing the dynamic stretching of the cable. The results show a fast-frequency component in the tension in the cable, representing the oscillations due to the dynamic stretching of the cable. Posiadała (1996, 1997) extended this model to a truck crane on an elastic support. The crane was modeled as a rigid body and the supports as elastic springs. The equations of motion were further augmented with six equations of motion describing the rigid-body motions of the

crane and boom. Their numerical simulations show that the free response of the payload is quasi-periodic. The slow frequency in the response (in-plane and out-of-plane pendulations) is due to the natural frequency of the cable–payload assembly, while the fast frequency is due to the support response and the resulting base excitations of the system at the boom tip.

Kościelny and Wojciech (1994) and Osiński and Wojciech (1994) modeled the unloading of cargo at sea. The crane and cargo ship were modeled as rigid bodies floating on the sea, the boom as a beam undergoing bending, the cable as a viscoelastic body, and the payload as a spherical pendulum. They wrote the equations of motion describing the planar motion of the system subject to nonlinear constraints representing the friction between the load and the cargo ship deck and contact between the boom and the crane ship.

Kral and Kreuzer (1995) and Kral et al. (1996), modeling a ship-mounted boom crane, augmented a planar pendulum model with the equations of motion of the ship modeled as a two-dimensional rigid body undergoing heave, surge, and pitch. The model was used to study the influence of cable hoisting on cargo pendulations. The results show nonlinear behavior, namely chaos at cable lengths exceeding 28 m and multiple responses at the same cable length, depending on the lowering or lifting history of the load.

Lewis et al. (1998, 1999) and Parker et al. (1999a, 1999b) modified the linear model of Parker et al. (1996) to represent a boom crane by replacing the trolley translation in the model with boom luff.

Towarek (1998) derived a model of a truck-mounted crane interacting with a flexible soil. The crane platform was modeled as a rigid body undergoing small oscillations, the boom as a flexible beam, the cable as an elastic string, the cable–payload assembly as a spherical pendulum, and the soil as a viscoelastic Kelvin–Voigt body. Using numerical simulations, they calculated the system response for complete revolutions of the boom slew at two different speeds. The system response shows that the crane oscillates with a narrow band of frequencies, thereby producing base excitations of the cable–payload assembly at the boom tip.

5. CONTROL

Management of payload motions varies according to the particular application at hand. In some applications, relatively large pendulations/oscillations are acceptable while the payload is en route to target destination as long as the *residual* pendulations/oscillations at the target point are small enough to allow for accurate payload positioning. In other applications, for example in a cluttered workspace or a nuclear reactor, more stringent conditions are imposed, requiring suppression of pendulations/oscillations along the travel path and at the target point. Considering the fact that most payloads are heavy, payload pendulations pose a safety hazard to workers and objects in the workspace and the structural integrity of the crane, thus more stringent motion suppression requirements are the norm rather than the exception.

These unwanted motions can arise as a result of inertia forces (due to the prescribed motion of the payload itself), base excitations (due to motions of the supporting structure), and/or wind loads on the payload. To avoid inertia-induced excitation, the crane operator has to perform maneuvers slowly. Furthermore, whenever unwanted payload motions develop, the operator has to either cease operations until the motion dampens out or perform counter-maneuvers to dampen out the motion. These constraints degrade the efficiency of crane operations and lead to time delays and high operating costs. Automatic control has gained

wide interest and application as a tool to alleviate, mitigate, or manage this problem. The increase in the payload weight, height, span, and speed of newly designed cranes necessitates even more effective controllers.

In the following, we discuss the crane control strategies and designs available in the published literature. Because of the structural difference between crane types, each type is discussed separately.

5.1. Gantry Cranes

Most control strategies designed for this class of crane assume a planar gantry crane, utilize planar, linear models, and assume that the crane path, external forces, and control effort are all planar. The forces they consider are exclusively inertia forces due to the acceleration and deceleration of the trolley, and the control effort they apply is the force or torque driving the trolley. This approach leaves the crane vulnerable to out-of-plane disturbances and the nonlinear coupling between in-plane and out-of-plane motions. Crane control strategies that take into account out-of-plane contributions of any of these factors or nonlinearities in the model are an exception rather than the rule. We will note these exceptions wherever they exist in the strategies and designs reviewed below.

5.1.1. Open-Loop Techniques

Input-Shaping

The most advanced and practical crane controllers today are controllers based on an open-loop approach designed to automate and/or shorten the cycle time for gantry cranes operating along a *pre-defined* path. The most widely used of the open-loop control techniques is *input-shaping*. Controllers using various forms of input-shaping are incorporated into gantry cranes currently used in ports (Hubbell et al., 1992) and elsewhere. This technique is used to move a crane a set distance along a set path. It has also been used to design "slow-down" mechanisms (covering only the deceleration stage of trolley travel) to ensure residual pendulation-free stop at the end of the trolley travel or for inching maneuvers in tight workspaces and near the target point. In this technique, the acceleration profile of the trolley travel is designed to induce minimum payload pendulation during travel and to deliver the payload at the target point free of *residual* pendulation. By design, the technique allows at least one-half of a pendulation cycle or integer multiples during each of the acceleration and deceleration stages.

Alsop et al. (1965) were the first to propose a strategy to control payload pendulations. The controller accelerates the trolley in steps of constant acceleration then kills the acceleration when the payload reaches zero-pendulation angle (after multiples of half of the period) and lets the trolley coast at a constant travel speed along the path. The same procedure is replicated in the deceleration stage. Assuming two constant acceleration/deceleration steps and a linear approximation of the cable–payload period, Alsop et al. used an iterative procedure to calculate the acceleration profile of the trolley. Their results showed that, although there was no residual pendulation, pendulation angles were of the order of 10° during the acceleration and deceleration stages. Carbon (1976) employed one-step and two-step versions of this strategy to decelerate the trolley and eliminate residual pendulations in the design of commercial gantry cranes used in ship unloading.

Using this strategy, Alzinger and Brozovic (1983) demonstrated, via a numerical example, that a two-step acceleration profile results in significant reductions in the travel time over a one-step acceleration profile. They used the two-step acceleration profile to design commercial gantry cranes used in ship unloading. Testing on an actual crane has shown that the two-step acceleration profile can deliver both fast travel and damped payload pendulation at the target point. However, testing has also shown that significant payload pendulations, as much as $5°$, arise and persist due to any deviations from the prescribed acceleration profile.

Hazlerigg (1972) proposed another input-shaping strategy using a symmetric two-step constant acceleration/deceleration profile to move the trolley to the target point in a period of time equal to the period of the cable–payload assembly and eliminate residual pendulation. The size of each of the two steps is determined based on the travel distance, the maximum available acceleration, and the period of the cable–payload assembly. Experimental verifications of the strategy showed that it dampened payload pendulations, however its performance was highly sensitive to changes in the cable length. Kuntze and Strobel (1975) extended this strategy by introducing one or more steps of zero-acceleration into the acceleration profile (that is, constant travel speed intervals), thus relaxing the constraint on the optimal travel time to one period of the cable–payload assembly and allowing for a constraint on the maximum travel speed to accommodate the capability of the trolley motor. Numerical simulation of the strategy showed that it was very sensitive to disturbances and parameter variations.

Yamada et al. (1983) proposed an input-shaping strategy where the acceleration profile is generated using Pontryagin's maximum principle to achieve minimum transfer time and no residual pendulations at the target point. They generated an optimal profile and approximated it with a suboptimal profile, using one or two steps of constant acceleration/deceleration along the path, in order to simplify the control effort. The suboptimal profile was then used to generate a "data table" containing the acceleration profiles for a grid of initial payload angular positions and speeds. The strategy was applied to a scaled model of a gantry crane. Results showed that the payload pendulations at the target point were less than $1.5°$.

Jones and Petterson (1988) extended the work of Alsop et al. (1965) using a nonlinear approximation of the cable–payload period to generate an analytical expression for the duration of the coasting stage as a function of the amplitude and duration of the constant acceleration steps. This analytical expression is then used to generate a two-step acceleration profile. Numerical simulations using various acceleration profiles show that this technique was able to reduce the residual pendulation to $0.1°$ to $0.3°$. However, it was not able to dampen out initial disturbances of the payload and could even amplify them. Noakes et al. (1990) and Noakes and Jansen (1990, 1992) applied a one-step variation of this acceleration profile to an actual bidirectional crane using a constant cable length and performing a U-shaped maneuver. Test results matched those of the numerical simulations.

Dadone and VanLandingham (2001) generated a better approximation of the cable–payload period using the method of multiple scales. Using numerical simulation, they compared the residual pendulations due to one-step input-shaping strategies based on their nonlinear approximation, a simplified form of that approximation, and the linear approximation of the period. They found a significant enhancement, of as much as two orders of magnitude, in the performance of the nonlinear control strategies over the linear strategy. The enhanced performance of the nonlinear strategies was most pronounced for longer coasting distances and higher accelerations.

Starr (1985) used a symmetric two-step acceleration/deceleration profile to transport a suspended object with minimal pendulation. The duration of the constant acceleration steps is assumed to be negligible compared to the period of the cable–payload assembly. A linear approximation of the period of the payload is used to generate analytically the acceleration profile. Strip (1989) extended this work by employing a nonlinear approximation of the cable–payload period to generate one-step and two-step symmetric acceleration profiles.

Karnopp et al. (1992) proposed an input-shaping strategy based on cam design techniques. Given the distance of the desired trolley travel and the pendulation natural frequency, it produces a prescribed input position of the trolley to deliver the payload residual-pendulation free at the target position. However, the minimum travel time has to be at least $\frac{1}{2}n$ of the payload pendulation period, where $n = 3, 5, 7, \cdots$. Also, using the minimum travel time (that is, 150% of the payload pendulation period) results in significant intermediate pendulation; the amplitude of this pendulation is 0.096 of the travel distance divided by the cable length. While a longer travel time results in reduced pendulation, it leads to further delays in operations and introduces higher harmonics in the motion.

Kress et al. (1994) have shown analytically that input-shaping is equivalent to a notch filter applied to a general input signal and centered around the natural frequency of the cable–payload assembly. Based on that, they proposed a robust notch filter, a second-order notch filter, applied to the acceleration input. Numerical simulation and experimental verification of this strategy on an actual bidirectional crane, moving at an arbitrary step acceleration and changing cable length at a slow constant speed, showed that the strategy was able to suppress residual payload pendulation.

Optimal Control

The first to propose a control strategy to automate crane operation was Field (1961). He used an analogue computer to simulate the dynamics of an ore unloading crane. By trial and error, he produced optimum velocity profiles for the trolley and cable motion that minimize the travel time while avoiding obstacles along the path. The control strategy, however, did not attempt to regulate payload pendulations.

Beeston (1969) used Pontryagin's maximum principle to generate time-optimal trolley acceleration profiles designed to minimize the hoisting and travel time for a single set of initial crane parameters and various target points. The strategy used bang-bang control of the trolley and generated three switching points for each acceleration profile. He then used regression analysis to express each of the switching points in terms of the initial trolley and payload position and velocity. This control strategy also did not attempt to regulate payload pendulations.

Manson (1982) relaxed the restrictions of Hazlerigg (1972) control strategy on the travel time and generated a time-optimal acceleration profile using Pontryagin's maximum principle. The three switching points on the acceleration profile and the total time were evaluated as functions of the travel distance only where the cable length was assumed to be constant. However, these optimal solutions were not practical to apply and were only meant as a benchmark for the performance of other strategies.

Karihaloo and Parbery (1982) proposed a strategy to eliminate residual pendulations for a given travel time and distance using Pontryagin's maximum principle. The control input was expressed as a function of the travel time and the masses of the trolley and payload.

Sakawa and Shindo (1982) applied the optimal control scheme proposed by Sakawa et al. (1981) for a boom crane to a model of a gantry crane linearized around the payload equilibrium position. They divided the pre-defined payload path into three stages: hoisting-up, travel, and hoisting-down. The scheme was applied to each stage independently. Simulation results show that, although the payload reaches the target point of each stage free of residual pendulations, significant pendulations, as much as 7°, develop along the hoisting-up and down (first and third) stages.

Kimiaghalam et al. (1998c, 1999b) used genetic algorithms to solve the optimal control problem formulated by Sakawa and Shindo (1982). In numerical simulation, they achieved similar results at a shorter travel time; however, their controller calls for drive speeds higher than the constraints on these speeds.

Auernig and Troger (1987) used Pontryagin's maximum principle to minimize the transfer time for a gantry crane traveling and hoisting at constant speeds subject to constraints on the maximum speed of the trolley travel and the cable hoisting. The technique produced the profile of the force to be applied to the trolley and/or the hoisting cable to generate the required motion profile. Their calculations of optimal control profiles showed that the optimal path generated by the technique was not always superior to the performance of non-optimal cranes in use at European ports. Also, while the payload was free of residual pendulation, significant pendulations developed along the path.

Hämäläinen et al. (1995) divided a pre-defined crane path into five stages: reeling in-place, reeling and trolley acceleration, coasting at a constant travel velocity, unreeling and trolley deceleration, and unreeling in-place. They generated the velocity profiles of the trolley and cable hoisting in the acceleration and deceleration (second and fourth) stages by minimizing the energy demand on the motors using a nonlinear model of the crane. The time required for these stages was minimized by trial and error. Numerical simulation and experimental verification on a scaled crane model showed that there were no residual pendulations at the target point; however, pendulations of as much as 6° developed during travel. They also found that performance under this control strategy was faster and smoother (that is, contained less pendulations) than that of a skilled operator using the same crane.

Optimal control techniques and input-shaping techniques are limited by the fact that they are extremely sensitive to variations in the parameter values about the nominal values and changes in the initial conditions and external disturbances and that they require "highly accurate values of the system parameters" to achieve satisfactory system response (Zinober and Fuller, 1973; Virkkunen and Marttinen, 1988; Yoon et al., 1995). While a good design can minimize the controller's sensitivity to changes in the payload mass, it is much harder to alleviate the controller's sensitivity to changes in the cable length. In fact, Singhose et al. (1997) have shown that input-shaping techniques are sensitive to the pendulation natural frequency. As a result, they suffer significant degradation in crane maneuvers that involve hoisting.

While closed-loop control may be used to alleviate these problems in input-shaping techniques, it cannot be used with time-optimal control techniques because it can lead to the development of limit cycles (van de Ven, 1983). Furthermore, the use of closed-loop control in conjunction with either approach requires a very accurate plant model and cannot therefore offer significant improvements over open-loop control (Zinober and Yang, 1988).

All input-shaping techniques and most optimal control techniques assume an undamped crane. The unaccounted-for damping in the crane system means that the payload will not

come instantaneously to rest at the target position as the simplified model suggests, thus *producing* residual pendulations. Finally, all control strategies in this class (except for that of Hämäläinen et al.) use a bang-bang acceleration profile. This profile applies excessive stresses on the crane structure and is difficult to generate accurately using industrial motors.

5.1.2. Closed-Loop Techniques

While open-loop techniques are, by definition, designed to suppress pendulations due to inertia excitations, all available closed-loop techniques are *by design* restricted to counter inertia excitations only. In these control strategies, the control input is the force or torque applied to the trolley and girder motor (where available) in order to suppress pendulations due to the acceleration and deceleration of the trolley.

Linear Control

Hazlerigg (1972) was the first to propose a feedback control strategy. It employed a second-order lead compensator to dampen the payload pendulations. Experimental verifications of the strategy showed that, while it dampened the payload pendulations at the natural frequency of the cable-payload assembly, it introduced pendulations at higher frequencies.

Ohnishi et al. (1981) used a two-phase strategy to dampen payload pendulations. The first phase is a linear feedback controller designed to stabilize the payload around its equilibrium position. To bring the payload to a stop, the trolley decelerates in two stages. The first deceleration stage is a part of the feedback control phase. The second deceleration stage is an input-shaping technique used to bring the load to rest over the target point. The control strategy was implemented on an actual overhead crane in a cold strip mill. While the strategy was able to minimize the pendulation angles, they reported that the automated system was 30% slower than the manual system it was supposed to replace.

Ridout (1987, 1989) proposed a feedback controller using negative feedback of the trolley position and velocity and positive feedback of the pendulation angle to eliminate residual payload pendulations at a constant cable length. Tests of the controller on a scaled model delivered the payload with less than a 0.3° pendulation angle to the target point; however, pendulations of as much as 10° developed during travel. He also found out that the controller was insensitive to external disturbances, changes in the payload mass, and *small* changes in the cable length.

To avoid persistent residual pendulation at the target point encountered in optimal-time control (due to unmodeled forces and disturbances), Virkkunen and Marttinen (1988) and Vähä and Marttinen (1989) proposed a combined control strategy using the Yamada et al. (1983) acceleration profile to drive the trolley all the way until the load is close to the target point and then switching to linear quadratic regulator (LQR) control to eliminate residual pendulations at the target point. The strategy was implemented on a scaled model of a gantry crane and results showed that it was successful in suppressing residual pendulations.

Moustafa and Ebeid (1988) proposed a strategy to suppress pendulations in a bidirectional crane by controlling both the trolley and girder motors. The strategy calls for three reduced-order feedback controllers, one for each of the acceleration, coasting, and deceleration stages of the motions. The controllers are based on linearization of the crane model around a single payload equilibrium position in each stage. Numerical simulations, for both the trolley

travel and traversing motion at a constant cable length, have shown that this technique can dampen inertia disturbances due to these motions. However, there were significant transient pendulations, as large as $20°$, and whirling motions, as large as $110°$, associated with both acceleration and deceleration.

Vähä and Marttinen (1989) and Virkkunen et al. (1990) proposed a P-controller applied to the trolley position and the payload pendulation angle to eliminate residual pendulations in a crane operating at a constant cable length and low travel speeds. In numerical simulation and experimental verification on a scaled crane model, the controller was successful in eliminating pendulations at the target point and limiting transient pendulations during travel to less than $3.5°$. However, the controller was sensitive to the payload initial conditions and the travel distance.

Caron et al. (1989) used a one-step acceleration profile to generate reference trajectories designed to minimize payload pendulations assuming either a constant cable length or a variable cable length. A PI controller is then used to track that path. Numerical simulations showed good tracking of the reference path with minimal transient pendulations of $1.7°$. Grassin et al. (1991) used LQR to track both of these reference trajectories. Numerical simulation, using the variable cable length strategy, and experimental verification, using a scaled model of a crane and the constant cable length strategy, showed smooth operation and transient pendulations less than $3.5°$. However, neither control strategies were able to reject disturbances to the payload angular position.

Yoshida and Kawabe (1992) designed a saturation linear state feedback controller to perform pre-defined maneuvers. Although the controller speeds up the travel, it incurs much larger pendulations than those suffered by a traditional linear feedback controller.

Moustafa and Emara-Shabaik (1992) used the model of Ebeid et al. (1992) to design a PD controller using the voltages of the trolley and girder motors as control input. Numerical simulations representing trolley travel *only* have shown that the controller is effective in suppressing payload pendulations.

Moustafa (1994) designed a linear feedback controller to suppress pendulations due to trolley motion and cable hoisting using the trolley motor force. This technique was applied to a linearized *time-varying* model of the crane. Results of computer simulation show that the technique is effective in suppressing payload pendulations, but it can develop a static error in the trolley position.

Nguyen (1994) proposed a state feedback control strategy to hoist, stabilize, and deliver the payload. Two independent controllers are employed: one (employing gain variation with cable length change) to control the trolley position and payload pendulation and another to control the payload hoisting position. Experimental verification on a scaled model demonstrated good tracking of the crane position and the cable length, no residual pendulations, and good damping of external disturbances to the trolley position and payload pendulation angle. However, there were transient pendulations of as much as $12°$.

Yoon et al. (1995) proposed a combined control strategy in which the second acceleration step and the coasting stage in an input-shaping two-step acceleration profile are replaced with feedback of the payload angular velocity to dampen payload pendulations. The underlying concept is for the feedback controller to alleviate the sensitivity of the input-shaping technique to external disturbances and changes in the cable length. Numerical simulation and experimental verification on a scaled crane traveling at a low speed have shown that this strategy is more capable of rejecting disturbances and adapting to changes in

the cable length than pure input-shaping. However, it is unable to reject disturbances in the deceleration stage. Furthermore, the ability of the fixed feedback gain to adapt to changes in the cable length is limited, and thus the strategy is unable to eliminate residual pendulations at the target point.

Yu et al. (1995) used a perturbation technique, the method of averaging, to separate the slow and fast dynamics of a gantry crane model. Two independent PD controllers are then applied. The first is a slow-input controller applied to the trolley to maintain tracking of a pre-defined motion profile. The second is a fast-input controller to suppress payload pendulations. Due to the approach used to develop the model, the controller can be applied only when the payload mass is an order of magnitude larger than that of the trolley. Simulation results have shown that this control strategy can move the payload along a pre-defined path smoothly with a maximum in-travel pendulation angle of 5°.

Lee et al. (1997) proposed a strategy composed of a PI controller to track the trolley position and a PD controller to dampen payload pendulations using the motion of the trolley. The control strategy behaves as a notch filter centered around the cable–payload natural frequency. Experimental verification using a scaled crane model running at a constant cable length showed transient pendulation of 3° during the acceleration and deceleration stages but no residual pendulations at the target point. Furthermore, it has shown that the control strategy is not sensitive to changes in the payload mass because of the high gear-reduction ratio of the trolley motor. However, it has also shown that the controller damping during travel is low, thereby leaving the payload vulnerable to external disturbance-induced pendulations.

Lee (1997) refined this strategy by compensating for the load applied to the trolley due to payload pendulations, cascading a PI trolley velocity controller with the PI trolley position controller, and cascading a lag compensator with the PD controller to increase the dampening of payload pendulations. He applied identical versions of this control strategy independently to the in-plane and out-of-plane motions of a bidirectional crane. Experimental verification using a scaled model showed small transient pendulations, less than 1°, and no residual pendulation. The control strategy is also more effective in resisting external disturbances and offers faster damping to payload pendulations throughout motion. The results have also shown that the PD controller is sensitive to changes in the cable length, thereby requiring *adjustment* of the gain to optimize the performance.

Assuming a flexible cable and a payload mass of the same order of magnitude as that of the cable, Joshi and Rahn (1995), Martindale et al. (1995), and Rahn et al. (1999) developed a linear feedback controller (PDC) to move the trolley from rest to a desired position and stabilize the vibrations of the cable–payload assembly at the endpoint of the maneuver. The controller design was verified experimentally using a scaled model. The authors reported robust response to "wind loading and time-varying cable length."

Alli and Singh (1999) proposed an optimal feedback controller applied to both a model assuming a rigid cable and another model assuming a flexible cable. The controller parameters are optimized to minimize the integral over time of the product of time and the magnitude of the error. Computer simulations have shown good regulation of payload pendulations, however the inertia forces involved in the simulation are minimal.

It should be noted that the underlying linearized crane model used in all of these strategies develops significant errors as the system parameters change over time. In particular, linear control strategies are invariably tuned to counter the effects of the natural frequency of the cable–payload at a single cable length. As a result, they are sensitive to changes in the cable

length. Therefore, linear control imposes restrictions on raising and lowering the payload during motion and requires low operating speeds, thus imposing unrealistic constraints on crane operations. Burg et al. (1996), simulating a classical linear feedback controller based on pole placement, reported that the linear controller produces large pendulations at small travel distances and complete revolutions of the payload at larger travel distances.

Adaptive Control

To account for the sensitivity of their input-shaping strategy to initial disturbances, Kuntze and Strobel (1975) used a linear crane model to predict the payload and trolley motions, to modify the acceleration profile accordingly, and to absorb these disturbances. They also updated the acceleration profile during operation to account for parameter variations (changes in the cable length and payload mass). Numerical simulation and experimental verification, using a scaled model, have shown that the strategy can effectively reduce the travel time and eliminate the residual pendulations.

Ackermann (1980) proposed a robust gain scheduling scheme for a linear state feedback controller. The scheme is designed as a fall-back controller to be activated in case of sensor failure or large changes in the states. It schedules the feedback gains to restrict the linear system poles to a region of stability instead of specific stable points. This control scheme reduces the performance to cope with system emergencies and assure stability. However, the underlying linear state feedback controller calls for a control effort to be applied to the angular velocity of the payload, but does not propose a mechanism to apply it.

Hurteau and DeSantis (1983) proposed an adaptive control strategy applied to a linear state feedback controller to eliminate residual pendulations. The strategy uses a gain tuning module to choose the gains to tune a pole-placement routine to changes in the cable length. Marttinen (1989), Salminen et al. (1990), and Virkkunen et al. (1990) proposed a similar fixed-parameter gain tuning strategy and a time-varying parameter strategy (updated according to cable length changes over time) to adapt the controller for changes in the cable length. Both strategies were verified on a scaled model of a crane. The results have shown that residual pendulations persist at the target point, significant pendulations develop during travel of as much as $10°$, and in the case of the time-varying parameter strategy a steady-state error occurs in the trolley position. It has also shown that the fixed-parameter strategy, unlike the time-varying parameter strategy, is insensitive to measurement errors.

Corriga et al. (1998) applied LQR to a linear time-varying model of a crane hoisting the payload at a constant speed. This implicit gain-scheduling procedure produced a gain vector that was a function of the time-varying length of the cable. Simulation results have shown that, while this control strategy was effective in rejecting initial disturbances, it was excessively slow in approaching the target point and displayed a steady-state position error.

D'Andrea-Novel and Boustany (1991a) and Boustany and d'Andrea-Novel (1992) used adaptive control to extend the applicability of the nonlinear controller proposed by d'Andrea-Novel and Lévine (1989) to a wider range of payload masses for the same controller parameters. However, this control strategy is only locally stable.

Butler et al. (1991) proposed a control strategy consisting of a primary controller, employing classical feedback control, and an adaptive controller to account for the unmodeled dynamics neglected in the linear reference model used for the design of the primary controller. To account for the unmodeled dynamics, they chose an unmodeled dynamics transfer function to minimize the plant-model error. The control strategy was verified using a scaled model of

a crane. The results showed significant reduction in residual payload pendulations after a few cycles of trolley travel along a pre-defined path.

Nguyen and Laman (1995) proposed a control strategy comprised of three independent H^{∞} controllers, one for each of the trolley position, hoist position, and payload pendulation. In experimental verification using a scaled model of a crane, the strategy produced small steady-state errors in the tracking positions and good dampening of payload pendulation to external disturbances. However, the strategy performance degrades as the acceleration applied to the plant increases.

Lee (1998) refined the control strategy of Lee (1997) by introducing a PI controller to the hoisting motor to track the cable length and gain scheduling to adapt the fixed gains of the payload pendulation feedback controller to slow changes in the cable length. The gains for optimum damping at each cable length are found, then curve fitting is used to express the gains as functions of the cable length. These functions are used to update the feedback controller gains in real time as the cable length changes. Experimental verification using a scaled bidirectional crane model running at low travel, traversing, and hoisting speeds showed transient pendulations less than 2° and no residual pendulations and an ability to reject external disturbances.

Méndez et al. (1998, 1999) used neural networks to enhance the performance of a state feedback controller and to tune it online to changes in the cable length. Two neural networks are used to model the dynamics and to generate and adjust the gains applied independently to the states of *each* of the trolley and the payload. The neural networks use a LQR structure to find the optimal gains based on the current states at each time step. Numerical simulation and experimental verification on a scaled model have shown that this strategy can produce a smooth positioning of the trolley and suppress residual pendulations at low travel speeds.

Fuzzy Logic Control

Yasunobu and Hasegawa (1986, 1987) and Yasunobu et al. (1987) proposed a predictive fuzzy control strategy to minimize payload pendulations and travel time, while moving towards a target point and maneuvering to avoid obstacles along the path. The strategy breaks the crane operation into seven stages and decides which fuzzy control rule to use in each of them, based on simplified models of the trolley and payload motions. The control rules then employ feedback control to command the trolley motion and cable hoisting. Experimental verification of the strategy using both a scaled model and an actual crane has shown that the strategy is more effective and consistent in minimizing the travel time and payload pendulation and more accurate in stopping at the target point than most skilled operators.

Yamada et al. (1989) used the trolley acceleration as input to move a crane at a constant cable length and minimize residual pendulations. A fuzzy logic controller imitates the suboptimal acceleration profile generated by Yamada et al. (1983). Using a scaled model of a crane, they compared this strategy to the input-shaping strategy they proposed (Yamada et al., 1983). They found that, while the two strategies have comparable performance, the fuzzy logic strategy is more effective in disturbance rejection. Suzuki et al. (1993) proposed a similar approach that, in addition, is capable of suppressing pendulations along the travel path and changing the cable length to avoid obstacles along the path. Numerical simulations have shown that, at a low travel speed, this strategy can avoid obstacles and dampen pendulations along the path to less than 1°.

Kim and Kang (1993) derived two fuzzy models of the crane dynamics to generate the reference velocities of the trolley and cable and then employed two fuzzy controllers to track these velocities. The control strategy is designed to minimize the travel time and payload pendulations while avoiding obstacles along the travel path. Numerical simulations have shown that the performance of the control strategy is comparable to that of a skilled operator.

Itoh et al. (1993, 1995) proposed a control strategy *imitating* an input-shaping acceleration profile with one step of acceleration and two steps of deceleration to minimize residual pendulation and to improve the accuracy of trolley positioning at the target point. Under this strategy, the cable length is held constant throughout the motion. Experiments conducted on an actual crane have shown that this strategy is more effective in payload pendulation suppression than a skilled operator or an input-shaping strategy.

Nalley and Trabia (1994) proposed a distributed fuzzy logic control strategy to dampen the pendulations of a bidirectional gantry crane. They used two independent sets of fuzzy inference engines (FIEs). Each FIE set has its own rules: one FIE set tracks the desired position, while the other corrects for payload oscillations. Each set is composed of two FIEs, one for each of the two perpendicular planes of crane motion. The outputs of the two sets of engines are added to obtain the total control input to the motors of the trolley and girder. The controller is used to drive the crane along a path generated by an input-shaping strategy. Simulation results have shown good damping of the pendulations.

Yoon et al. (1995) proposed a fuzzy controller designed to emulate the acceleration profile in their combined strategy except in the deceleration stage where it emulates a target point position feedback. Numerical simulation and experimental verification on a scaled crane traveling at low speed have shown that the strategy can suppress residual pendulation and tolerate changes in the cable length away from the nominal value. However, external disturbances lead to oscillations of the trolley around the target position.

Liang and Koh (1997) used a fuzzy logic controller to eliminate residual pendulations at the target point using a heuristic approach to minimize pendulations. The trolley decelerates as it approaches the end point, thus producing inertia-induced pendulations. It then accelerates to bring the trolley directly above the payload when it reaches the maximum point on its upward swing and thus is temporarily at rest. This procedure is repeated until the payload is at rest. Computer simulations have shown that, even though a few cycles of this procedure can bring the payload to rest, significant pendulations develop in the process. Méndez et al. (1999) proposed a similar fuzzy controller employing the position of the trolley and the pendulation angle to eliminate residual pendulations. Experimental verification has shown that the fuzzy controller makes the trolley arrive at the target position smoothly with no residual pendulation; however, to achieve that it approaches the target point very slowly.

Kimiaghalam et al. (1998a, b) used the model of Sakawa and Shindo (1982) to design a fuzzy logic controller to move the payload from one side of a fixed obstacle to a known destination on the other side without collision and in a relatively short time. The controller imitates the human decision-making process. Two designs of the controller are proposed. The first produces torques as a function of the payload position, while the second generates desired speeds of the trolley and hoist from which torques are computed using a PD controller. Simulation results have shown that the first design is faster, while the second is relatively slower but yields a smoother path. Using the second design to achieve higher speeds produces larger pendulations and trolley oscillations around the target point.

Fuzzy logic strategies are especially hard to tune. The control input is either too high, which produces cycles of overshoot–undershoot around the target point, or too low, which produces a very slow and *time-consuming* approach to the target point. Furthermore, all strategies in the literature restrict crane operation to a pre-defined path.

Nonlinear Control

Zinober (1979) proposed a sliding-mode control strategy to minimize the travel time, to eliminate residual pendulations, and to avoid obstacles along the travel path. The strategy is not a function of the crane parameters and thus is not sensitive to changes in the cable length and payload mass. It employs a linear switching function of the system states to switch up and down a bang-bang controller of the torque applied to the trolley. A low-pass filter is then applied to the control input to eliminate high-frequency components from the input signal. Numerical simulations have shown that the travel time is 10% longer than the optimal travel time, however the strategy is able to reject external disturbances without degrading the system performance.

D'Andrea-Novel and Lévine (1989) have shown that *static* state feedback linearization works only when starting from a stable configuration and moving at slow rates and even then can only ensure local stability. On the other hand, *dynamic* state feedback linearization can stabilize the system for any initial configuration and for higher speeds. They demonstrated this result on a crane traveling and hoisting along a pre-defined path at constant speeds. However, their controller is dependent on payload mass.

Fliess et al. (1991, 1993) proposed a nonlinear *dynamic* state feedback technique to linearize the dynamics of a crane. The technique, dubbed flatness-based control, is applicable to *flat* systems only, that is systems where the input and state variables can be expressed in terms of the output variable and their time derivatives in closed form. Thus, based on inverse dynamics analysis of the nonlinear planar model, they write the system inputs, hoisting and traversing accelerations, in terms of the system outputs, payload position. Substituting the mathematical representation of the desired trajectory into these nonlinear expressions produces the required input accelerations and results in a linear relationship between the state and input variables. A PI controller is then used to drive the trolley and hoist motors to track these pre-defined input accelerations. Computer simulations have shown an enhanced performance in the trolley and the payload positioning tasks with improved operation time. Payload pendulations were reduced to a maximum of 1.7° during the maneuver.

Bourdache-Siguerdidjane (1993, 1995) applied a variation of the Fliess linearization technique to an extended model, including the dynamics of the trolley. The nonlinear model is first linearized by matching it to a version of the model linearized around a single equilibrium. LQR is then applied to the new linear system to generate the gains of the feedback controller, which drives the motors and tracks the reference payload path. Simulation results have shown that this strategy eliminates the payload residual pendulation at the target point.

Maier and Woernle (1997) applied yet another variation of the Fliess linearization technique to an extended model including the dynamics of the trolley and the hoisting motor. First, inverse dynamics are used to linearize the model. Then cascaded feedback control using pole-placement is applied first to the payload position and then to the trolley and hoisting motor positions in the linear system to counteract the effect of external disturbances and unmodeled forces. Simulation results have shown that *this* application of the Fliess linearization technique is capable of rejecting initial disturbances to the payload position.

This control strategy, however, requires an exceptionally *smooth* trajectory to produce practical inputs because the inputs are functions of the trajectory and its time derivative up to and including the fourth-order derivative.

DeSantis and Krau (1994) proposed a sliding-mode control strategy to stabilize in-plane and out-of-plane pendulations of a bidirectional crane. First, two independent, planar state feedback controllers estimate the control input of each motor in order to suppress inertia-induced payload pendulations. Sliding-mode control is then applied to these estimates to produce the actual control input of the motors. In numerical simulations, they compared the sliding mode to plain state feedback control strategies and found out that both approaches are able to stabilize the payload motion at low trolley and girder speeds. However, sliding-mode control is more effective in coping with changes in the crane parameters (payload mass) and external disturbances and less effective in handling feedback delays, as *compared* to plain state feedback control.

Martindale et al. (1995) proposed two feedback control strategies to track a pre-defined path. The first applies backstepping control, and the second adds an adaptive gain matrix to the controller to account for uncertainty in the model parameters (trolley mass, payload mass, and viscous damping applied to the trolley). Experimental verification using a scaled crane model has shown that both control strategies have the capacity to suppress pendulations at low trolley travel speeds. It should be noted, however, that backstepping-like flatness-based control uses the fourth-order derivatives of the output, thereby requiring a smooth trajectory. Burg et al. (1996) used the Teel saturation control approach to design a third feedback controller to minimize payload pendulations. Experimental verification using the scaled crane model has shown that, while the controller has a capacity to suppress pendulations at low trolley speeds, significant payload pendulations develop at higher speeds.

Cheng and Chen (1996) proposed a control strategy which employs feedback linearization and time delay control to move a crane along a pre-defined *smooth* path and to eliminate residual pendulations. Numerical simulation has shown that the strategy is able to deliver the payload with no residual pendulation and minimal transient pendulation, less than 3°. Furthermore, unlike pure feedback linearization, it is robust enough to handle changes in the payload mass and unmodeled forces. Its performance is also better than the adaptive feedback linearization of d'Andrea-Novel and Boustany (1991a) and Boustany and d'Andrea-Novel (1992) since it does not overshoot the target point. All three feedback linearization approaches are sensitive to external disturbances, which increase transient pendulations significantly.

Assuming a flexible cable and a payload mass of the same order of magnitude as that of the cable, d'Andrea-Novel et al. (1990, 1994) and d'Andrea-Novel and Boustany (1991b) proposed two embodiments of a feedback controller. In one, the dynamics of the trolley are ignored and a nonlinear feedback law utilizing the trolley speed is proven to be able to uniformly stabilize the cable–payload assembly. In the other embodiment, the dynamics of the trolley are included in the model and a nonlinear feedback law utilizing the force applied to the trolley is proven to be able to stabilize the cable–payload assembly. However, the stabilization of the system in this case is not uniform. Computer simulations using a *linear* feedback law were used to demonstrate both strategies. The results indicate successful stabilization of the cable–payload assembly.

5.2. Rotary Cranes

Rotary and boom cranes are seldom used to perform planar tasks. As a result, most control strategies proposed for both crane classes are three-dimensional. On the other hand, the few rotary crane-control strategies available in the literature deal exclusively with inertia-induced payload pendulations even though base excitations are possible and wind-gust excitations are probable in the operation of rotary cranes. Furthermore, it should be noted that stabilizing the payload against the translational motions of a gantry crane is less complicated than stabilizing the payload against inertia-induced pendulations of the slew motion in rotary and boom cranes, which produces pendulations in both the radial and tangential directions.

Gustafsson (1995) and Abdel-Rahman and Nayfeh (2001) have shown that a single planar controller cannot stabilize the payload against slew-induced pendulations in a boom crane. Using an out-of-plane controller only, Gustafsson (1995) was able to stabilize the out-of-plane motion of the payload, but could not stabilize the in-plane motion. Thus, he concluded that the control effort had to be applied both in-plane and out-of-plane to completely stabilize the payload.

5.2.1. Open-Loop Techniques

Parker et al. (1995a,b) applied various optimization techniques to input-shaping of the acceleration profile of the jib in order to eliminate residual pendulations for a jib maneuver along a pre-defined path. Experimental verification has shown that significant pendulations develop during the maneuver, reaching as much as 10° for the given maneuver.

Parker et al. (1996) presented another control strategy to drive both the trolley and the jib. It uses a quasi-static notch filter to eliminate excitations of the cable–payload assembly at the natural frequency from the slew and travel inputs of the operator. The notch location varies with the length of the cable to filter out excitations at the current natural frequency of the cable–payload assembly. The roll-off coefficient for the notch filter is held constant and thus is optimum only at a single cable length, and the filter characteristics change with changes in the cable length. Experimental results have shown a significant reduction in the payload pendulations throughout the maneuver. However, the filtering process imposes delays, as much as 2.5 s, between the operator input and the actual filtered input to the crane. Furthermore, the variable filter characteristics produce variable responses for the same operator input. Also, the linear nature of the filter limits its effectiveness to low crane speeds.

5.2.2. Closed-Loop Techniques

Golafshani and Aplevich (1995) used a time-optimal control scheme to generate trajectories of the jib, the trolley, and the cable length. A sliding-mode controller is then used to track these trajectories. In computer simulations, the time-optimal trajectories produced uncontrolled payload pendulations. The constraint on the time was therefore relaxed to 110% of the optimal value, and suboptimal trajectories satisfying a minimum payload swing energy condition were then used instead of the optimal trajectories. Computer simulations have shown that the suboptimal trajectories reduce the payload pendulations. However, significant pendulations persist throughout the maneuver and at the end point.

Almousa et al. (2001) used two FIEs, one for the motion in the radial direction and another for the motion in the tangential direction, to track the position of the payload and two other FIEs, one for each of the radial and tangential directions, to dampen the payload pendulations. Each of the FIEs operates independently from the others. Computer simulations have shown that the fuzzy logic controller can limit in-plane and out-of-plane pendulations to small angles throughout jib and trolley maneuvers. It can also dampen pendulations due to disturbances to the trolley and jib positions. However, the control strategy imposes an increase in the crane maneuver time.

Omar and Nayfeh (2001) applied two full state feedback controllers independently to the trolley travel and the jib slew. This control strategy was effective in damping payload pendulations within one cycle of oscillation but only when the feedback gains were tuned for a specific payload mass and cable length. Changes in these parameters led to marked degradation in the controller efficiency.

5.3. Boom Cranes

The prediction and control of payload motions in boom cranes is more complicated than it is for other types of cranes because of the coupling between the payload response to the slew, luff, and hoisting motions. Furthermore, because of the mobile nature of most boom cranes, it is impossible to isolate the crane from base excitations. Consequently, any effective control strategy for boom cranes has to account for base excitations. On the other hand, it is not necessary to account for the crane structure elasticity in the base excitations. Osiński and Wojciech (1998) used nonlinear optimization to generate an input-shaping profile of either the moment or the velocity of the hoist motor during the lifting of a load off a cargo ship by a boom crane. To model the plant, they simplified the model of Kościelny and Wojciech (1994) and Osiński and Wojciech (1994) by assuming an immobile crane ship and reducing the sea effect on the cargo ship to a harmonic heave motion. They found that including the elasticity of the boom had "only a small influence on load motion."

To provide the control authority necessary to suppress base excitations, some researchers found it necessary to use a variety of specially constructed add-on actuators, in addition to the boom slew and luff actuators. This approach, however, makes these control strategies more expensive and cumbersome to use.

5.3.1. Open-Loop Techniques

Sakawa et al. (1981) proposed an optimization scheme to generate the torque profile necessary to transfer a load along a pre-defined path while minimizing the payload pendulations during transfer and at the target point. The transfer time is minimized by iteration. The technique was applied to a simulated model of a boom crane slewing at a *constant* luff angle while reeling in the cable and linearized around the payload equilibrium position. Simulation results have shown that the payload is free of residual pendulations at the target point, however payload pendulations develop along the path and increase as the slewing angle increases.

Takeuchi et al. (1988) proposed an input-shaping strategy to achieve a time-optimal slew motion *only* while minimizing the residual pendulations. The strategy uses a slew angle acceleration profile similar to that generated by Yamada et al. (1983), for gantry cranes, to

perform the slew motion and control the pendulations. Numerical simulations have shown that the strategy can suppress out-of-plane pendulations but not in-plane pendulations, which persist well after the boom comes to a stop.

Lewis et al. (1998) applied the control strategy proposed by Parker et al. (1996) to boom cranes. Simulation results have shown reductions in both the in-plane and out-of-plane payload pendulations. However, the control strategy in this case has the same limitations observed when applied to rotary cranes. Parker et al. (1999a, b) modified this control strategy using a roll-off coefficient linearized with respect to the natural frequency employed in the notch filter design. Experimental verification showed an 18 dB reduction in the payload pendulations at the end of the prescribed maneuver. Simulation results (Lewis et al. 1999) have shown that, while the response of this filter is more consistent at different cable lengths and demonstrates a slight improvement in the pendulation reduction over that of Parker et al. (1996), it imposes more time delays and larger changes of amplitude on the operator input. Alternatively, Lewis et al. (1999) modified the same control strategy using a roll-off coefficient linearized with respect to the forcing (slew and luff input velocities) in the notch filter. Simulation results did not show a significant difference in the performance of this filter as compared to that of Parker et al. (1999a,b).

5.3.2. Closed-Loop Techniques

Sakawa and Nakazumi (1985) proposed a two-tier control strategy for a crane traversing a pre-defined trajectory. An open-loop controller tracks the trajectory the boom travels, while a LQR optimized state feedback controller employs the slew, luff, and hoisting to eliminate inertia-induced residual pendulations at the end of the maneuver. Computer simulations have shown pendulation angles during the maneuver of as high as 21.6°. Sato and Sakawa (1988) extended the application of this control strategy to a boom with a flexible jib at the boom tip.

Takeuchi et al. (1988) proposed a fuzzy logic strategy to achieve a time-optimal slew motion *only* while minimizing the residual pendulations. The strategy imitates the input-shaping strategy proposed in the same work. Numerical simulations have shown that the fuzzy logic controller is unsuccessful in dampening in-plane pendulations. This is expected since the control effort applied, the slew motion, is an out-of-plane motion.

Hara et al. (1989) proposed a LQR optimized state feedback controller using the boom telescopic motion as a control input to control *planar* payload pendulations due to the telescopic motion of the boom. A saturation condition is applied to the controller input to the plant to keep it within available control authority, thus producing a suboptimal controller. In computer simulations and testing on an actual crane, the control strategy was successful in suppressing pendulation.

Nguyen et al. (1992) proposed a state feedback control strategy, based on a linear *planar* model, to hoist and stabilize the payload and position the boom. Two independent controllers are employed, one to control the boom luff angle and payload pendulation and another to control the payload hoisting. Experimental verification on a scaled model showed oscillations of the boom around the reference path and steady-state errors in the boom angle and cable length. On the other hand, transient pendulations were contained to less than 4°.

Souissi and Koivo (1992) proposed a two-tier control strategy to stabilize a boom crane against *inertia-induced* payload pendulations. A PID controller tracks a reference trajectory using the slew and luff of the boom and the reeling/unreeling of the cable, while a PD

controller dampens the payload pendulations. Numerical simulation of the boom performing a pre-defined luffing–slewing–luffing maneuver at a constant cable length showed significant payload pendulations, as much as $15°$, indicating that the PD controller was not effective in damping the pendulations.

Gustafsson (1995) proposed a control strategy employing two independent, in-plane and out-of-plane, linear position feedback controllers designed based on a partial linearization of the spherical pendulum model to suppress *inertia-induced* payload pendulations. Computer simulation results showed stable responses for operator commanded slewing rates away from the natural frequency of the cable–payload assembly and small pendulation angles.

Chin et al. (1998) proposed a nonlinear feedback control scheme to suppress the parametric instabilities in payload motions due to wave-induced base excitations. They demonstrated analytically that introducing a control effort in the form of a harmonic change in the cable length at the same frequency as the base excitations can suppress the parametric instability and result in a smooth response.

Abdel-Rahman and Nayfeh (2000) used cable reeling/unreeling to avoid whirling motions and three-dimensional responses of the payload when the frequencies of the base excitations approach the natural frequency of the cable–payload assembly. Using numerical simulation, they have demonstrated that the scheme changes the underlying dynamics of the payload motion, allowing the primary controller to exert an effort to dampen a planar motion instead of attempting to dampen a three-dimensional harmonic or chaotic motion.

Using a *planar* model of a ship-mounted crane, Henry et al. (2001) developed a delayed feedback controller. Computer simulations and experimental results showed an effective suppression of payload pendulations due to in-plane, roll and heave, excitations. Masoud et al. (2000) extended this approach to the three-dimensional case. In computer simulations and experiments, the controller successfully suppressed payload pendulations due to both in-plane and out-of-plane base excitations.

5.3.3. Control Strategies Employing Modifications of Crane Structure

The most basic and the only system in practical use of this class is the Rider Block Tagline System (RBTS), Figure 7. Under this design, a rider block is attached to the cable. Using two pulleys and taglines, the crane operator can pull the block towards the boom and move it up and down the cable, thereby decreasing the effective length of the cable and increasing the natural frequency of the cable–payload assembly (Bostelman and Goodwin, 1999). This process is used to detune the natural frequency of the cable-payload away from the excitation frequency. In practice, this approach has proven to be cumbersome; the taglines tend to entangle with the cable or jump their own drum, thereby necessitating re-rigging of the crane.

Lévine et al. (1997) modeled a boom crane equipped with a single pulley and tagline allowing the operator to pull the cable towards the boom *only*. The model augments the planar pendulum equation of motion with two equations of motion describing the torques of the payload and tagline motors. It assumes viscous damping in both motors and a massless pulley. Lévine et al. (1997) have proposed an open-loop controller where flatness-based control is used to generate the torque of the two motors from the payload trajectory described by a *smooth* curve.

Ott et al. (1996) and Yuan et al. (1997) proposed a system to rig ship-mounted cranes dubbed “Maryland rigging”, Figure 8. Under this rigging system, the payload suspension



Figure 7. A boom crane equipped with the Rider Block Tagline System.

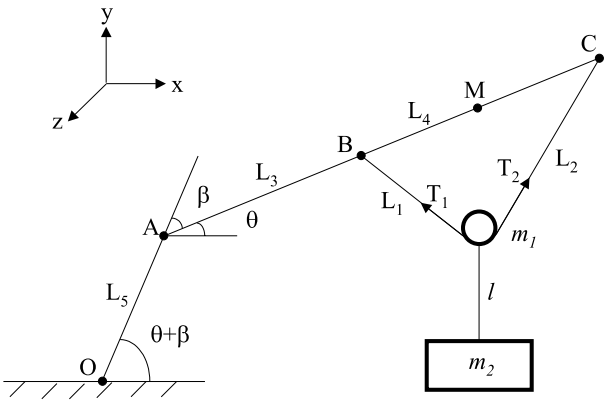


Figure 8. Schematic diagram of a boom crane equipped with Maryland rigging.

is transformed from a single spherical pendulum to a double pendulum system. The upper pendulum is a pulley riding on a cable suspended from two points on the boom; the pulley is thus constrained to move over an ellipsoid. The lower pendulum is the payload suspended by a cable from the pulley. It continues to act as a spherical pendulum. A passive control effort is applied to the *planar* payload pendulations by applying a brake system to the upper cable as it passes through the pulley. Yuan et al. (1997) derived a planar model of this rigging and used it to investigate the system response to periodic and chaotic roll motions. Their simulation results have shown that the payload response grows significantly when the period (dominant period in the chaotic motion case) approaches the natural frequency of the lower pendulum. The pulley was then used as a brake to apply a constant and continuous dry (Coulomb) friction. Simulation results have shown that a constant friction at a level equivalent to 10% of the payload weight can reduce *planar* payload pendulations significantly even in the neighborhood of the natural frequency.

Kimiaghalam et al. (1998d) used an FIE to determine the level of Coulomb friction in the brake of the Maryland rigging. Simulation results have shown that the performance of the active friction control is comparable to that of the original passive friction control, while the required control effort is decreased. Kimiaghalam et al. (1999a) proposed another fuzzy logic control approach to dampen the pendulations in a Maryland rigged crane. The FIE does not apply any friction through the pulley, instead it changes the upper cable length to eliminate the pendulations. Simulation results have shown that this control strategy can dampen payload pendulations; however, its performance was inferior to that of the passive controller.

Dadone and VanLandingham (1999) proposed a combined control strategy to stabilize the in-plane motions of the payload under the Maryland rigging. A fuzzy logic inference engine is used to determine the level of dry friction in the pulley based on the positions and velocities of the pulley and payload. Simultaneously, the pulley velocity and acceleration are used to feedback changes in the pulley cable to eliminate vertical oscillations of the pulley. Simulation results have shown fast damping of the payload motions; however, the friction level (control effort) in the pulley, at 40% of the payload weight, was much higher than that employed by Yuan et al. (1997). Furthermore, the changes in the length of the pulley cable absorbed the pulley vertical oscillations, but also it introduced horizontal oscillations in the positions of both of the pulley and payload.

Wen et al. (1999) and Kimiaghalam et al. (2000b) proposed a combined feedforward and feedback control strategy to stabilize *planar* pendulations in a crane equipped with the Maryland rigging. The feedforward law is based on the linearized planar equations of motion. It changes the upper cable length to cancel the effects of the base excitation due to ship roll. The feedback controller applies LQR feedback control to changes in the upper cable length to add damping to the system. Simulation results have shown that the controller can reduce the payload pendulations to less than 3° for small roll motions. Kimiaghalam et al. (2000a) proposed another combined feedforward and feedback control strategy. The feedforward controller uses boom luffing to reduce the excursions of the equilibrium point of the pulley due to ship rolling. The feedback controller changes the upper cable length to keep the pulley positioned directly above the payload as the ship rolls. Simulation results have shown that the combined control strategy is both effective and fast in suppressing payload pendulations due to both ship roll and initial disturbances. However, both feedback controllers assume full authority over the lengths of both segments of the upper cable, and hence the pulley position. This assumption violates the pulley's equilibrium equation.

Abdel-Rahman and Nayfeh (2001) have shown that the nonlinear coupling between the in-plane and out-of-plane motions of traditionally rigged cranes continue to exist in Maryland rigged cranes, leading to out-of-plane motion due to in-plane excitation and jumps in the in-plane motion, thereby suggesting that a planar control effort cannot stabilize the payload motions. Simulation results have shown that, while the control mechanism was successful in limiting the in-plane motion, it was unable to control the out-of-plane motion and could not stabilize the overall payload motions.

Balachandran and Li (1997) and Balachandran et al. (1999) used two-dimensional and three-dimensional (Li and Balachandran, 1999) nonlinear models of ship-mounted boom cranes to design a nonlinear vibration absorber, a mechanical filter, to control the motions of the pivot point around which the payload oscillates. The design modifies the boom crane configuration to suspend the payload from a pivot plane, which in turn is suspended under the boom tip. The absorber has both a passive mode and an active mode employing feedback control. Computer simulations have shown that the absorber can shift bifurcation points arising from the nonlinear dynamics of the cable–payload assembly and suppress subcritical bifurcations. This has also shown that the feedback component of the filter can attenuate the transient and steady-state payload motions. However, in some filter designs, suppression of the subcritical bifurcation produced a large resonance-like response around half the natural frequency.

Iwasaki et al. (1997) and Imazeki et al. (1998) designed an active mass-damper system to suppress payload pendulations. The system was installed on the sling of a barge-mounted boom crane. A planar linear model of the crane was used to design a linear feedback controller. The control effort is the acceleration applied to a 35-ton damping mass riding on the sling. A 132 KW induction motor is used to drive the damping mass. Test results showed that the sling motion was reduced to $\frac{1}{2}$ to $\frac{1}{3}$ of the uncontrolled motions at the test frequency.

Dadone et al. (2001) used a variable-geometry truss (VGT) to suppress in-plane payload pendulations of a ship-mounted boom crane undergoing roll and heave excitations. Actuators embedded in the VGT apply an acceleration control effort to a control point on the cable and constrain it to move along a straight line. LQR and fuzzy logic procedures using the positions and velocities of the payload and the control point were designed to minimize the control effort based on a linear, planar model of the modified crane. Simulations showed that the fuzzy logic version of this control strategy was effective in suppressing payload pendulations throughout the bandwidth of excitation frequencies, while the LQR version was effective only where the system behavior was almost linear; i.e. at low excitation frequencies *away* from the natural frequencies. Comparison of the two versions of the strategy showed that the fuzzy logic version applied larger control effort than the LQR version.

6. SUMMARY

6.1. State-of-the-Art

A significant research effort has been devoted over the past 25 years to the development of control strategies to improve the efficiency and safety of cranes. Most of this research has been limited to addressing inertia-induced pendulations in gantry cranes operating along a pre-defined path. Input-shaping techniques demonstrated a potential for increasing the

hoisting, travel, and traversing speeds of gantry cranes. However, they are not robust enough to reject external disturbances or stabilize the payload under base excitations and unmodeled forces in the plant. As a result, they are not able to relax the operability constraints.

Linear control techniques were added to input-shaping based strategies to alleviate these shortcomings. However, they are not robust enough to allow for variations in the cable length and payload mass, high operating speeds, and large changes in the trolley and payload positions. Fuzzy logic and adaptive control techniques were also used to supplement input-shaping techniques. While hybrid techniques have the potential to produce robust and efficient control strategies, experience until now shows that the design of control strategies using these techniques is not trivial.

Research on boom and rotary cranes is still in the preliminary stages, as compared to research on gantry cranes where some of the proposed strategies have been put to work in the field. However, even in this case, most of these control mechanisms have proven to be inefficient and thus have been "locked out and abandoned" by the operators (Hubbell et al., 1992). The only exception to that are input-shaping *based* controllers. They are used both to operate cranes as well as to perform operator initiated short steps (so-called *inching*) used in precision maneuvering near the target point and around obstacles. The recent availability of variable-speed AC (flux vector) drives has made generation of the bang-bang acceleration profiles, typical of input-shaping strategies, feasible as demonstrated by Noakes et al. (1993) and Kress et al. (1994) on an actual gantry crane, thus removing one of the main hurdles to practical implementation of this class of control strategies. However, input-shaping strategies continue to be hobbled by pendulations along the travel path, relatively slow speeds, and the fact that most of them are designed for a particular cable length, and thus their behaviors at other cable lengths are suboptimal. They, also, are not effective in disturbance rejection and are sensitive to unmodeled forces, such as friction. Fuzzy logic and adaptive control strategies based on input-shaping have proven to be more effective in disturbance rejection and less sensitive to unmodeled forces and parameter variations than plain input-shaping.

At the root of the mismatch between the large body of research on crane controllers and those in practical use is another mismatch between the focus of research effort and operators' interests. Most of the research work has been directed to crane automation. On the other hand, operators in the field are not interested in full automation because of concerns about controller robustness, safety restrictions, or a workplace that mandates a flexible crane allowing for a variable trajectory from one operation cycle to the next.

6.2. Modeling

The complexities of dealing with a nonlinear model of the plant drive most of the work on crane control to make do with linearized approximations of the model. This simplification comes at the price of reduced controller robustness. Burg et al. (1996) reported that the neglected nonlinearities in a state-space model of a gantry crane may significantly impact the performance of a linear controller. Their computer simulations have shown that a linear controller provides acceptable performance only within a fixed operating range of small pendulation angles around the equilibrium point of the payload. As a result, there has been an increasing interest in the design of crane control strategies based on nonlinear crane models.

As demonstrated in the analysis of the spherical pendulum model and indicated by Gustafsson (1995) and Abdel-Rahman and Nayfeh (2001), the in-plane and out-of-plane

motions of the payload are coupled. Motion in one plane interacts with and induces motion in the other plane. Whenever large pendulations build up in-plane, any out-of-plane disturbance can give rise to out-of-plane pendulations, and thus the planar model breaks down. Therefore, three-dimensional nonlinear models, accounting for both geometric and kinetic nonlinearities in the payload motion, must be adopted as the gold standard of the field.

Simplified models are still a good approximation, they are legitimate and useful under special loading conditions. A planar model, equations (45) or (47), can be used to model a gantry crane as long as the payload and trolley are not subject to any large and/or out-of-plane excitations (wind gusts, load imbalance, or girder deviation) and the safety threshold in the particular application is low. The nonlinear planar model, equation (47), however is superior to the linear model, equation (45), since it is consistent and valid for small but finite pendulation amplitudes. Linear models can also be used as long as the pendulation angles are small and the frequencies of all present base excitations are away from the natural frequency of the cable–payload assembly.

A recent development in gantry cranes has been the introduction of multiple-point payload suspension. This design allows for enhanced stiffness of the cable–payload system and thus more resistance to pendulations. Furthermore, new control schemes have been introduced to use these multiple points of suspension to dampen payload pendulation using the differential between the tension forces in the various suspension cables (Champion, 1989; Hubbell et al., 1992). This control approach mandates a model to account for the payload motion as a *rigid body* rather than a point mass. To date, we are not aware of any such model in the literature. Furthermore, this class of control strategies is not effective in disturbance rejection. Wind gusts and/or initial disturbances induce payload pendulations, which cannot be stabilized with these controllers.

6.3. Control

Despite the numerous crane control strategies in the literature, very few designs have proven applicable in practice. One reason is that most of these strategies were not designed with a crane in mind. In many cases, a crane was being used as a test-bed for a novel control concept. We propose that a successful crane control strategy has to meet the following criteria:

- The advantages of using a crane over a robotic arm or a multiple-winch crane are flexibility, cost efficiency, and simplicity of design and operation. Any crane control strategy has to maintain these advantages. An appropriate guideline is for control strategies to utilize available actuators within their existing power limitations.
- Most cranes are manually operated to maximize the flexibility, robustness, and safety of crane operation. A control strategy designed for this class of crane will have to be transparent to the operator, thus precluding any control strategies that result in significant delays in response to the operator input.
- The use of automated cranes is mostly limited to mines, factories, and similar installation where a material handling system is required to reproduce a set sequence of motions. Automating a crane is only feasible where the workspace is well structured with constant starting and target points, fixed positions of obstacles along the crane path, and a low safety threshold. Furthermore, the control strategy employed in automating the crane will have to be robust enough to handle a wide range of cable lengths and payload weights.

- The light damping of cranes means that control strategies must be designed to take into account both stationary and transient responses.
- The control strategy has to apply control effort in two perpendicular planes. Planar controllers can only be used in the absence of nonplanar excitations. Even in a unidirectional gantry crane, such a condition is not guaranteed due to load imbalance, girder deviation, and wind gusts. The only exception under which planar crane control is safe is where the three-dimensional motion is modeled and a safety mechanism/controller is incorporated to bring the payload motion back from whirling to a planar motion.

For the same class of models, the performance of nonlinear controllers shows dramatic improvement over that of linear controllers.

Acknowledgment. This work was supported by the Office of Naval Research under Grant No N00014-96-1-1123 (MURI).

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